Design and Development of IoT-Based Smart Maldive Fish Processing System

INDIVIDUAL THESIS

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ABSTRACT

In traditional Maldive fish (*Umbalakada*) processing, raw tuna intake quality is evaluated using subjective manual organoleptic inspection. This manual approach fails to detect early internal proteolysis and microbial decay prior to superficial discoloration, resulting in inconsistent batch quality and processing losses exceeding 20%. This research addresses this gap by developing an automated, non-destructive, multimodal Artificial Intelligence (AI) raw fish quality assessment system for real-time inline intake inspection. The framework fuses an edge-optimized YOLO11s vision model with a calibrated MQ-135 electrochemical metal-oxide semiconductor gas sensor array positioned within an enclosed sampling hood above an intake conveyor. The system classifies raw skipjack tuna (*Katsuwonus pelamis*) into three freshness tiers: *Very Fresh*, *Fresh*, and *Spoiled* using a late-fusion gating rule that combines visual confidence vectors with headspace gas readings.

Evaluated across a dataset of 2,536 annotated skipjack tuna frames, the FP16-quantized YOLO11s model achieved an overall mAP@0.5 of 95.8%, precision of 94.1%, and recall of 93.7%, executing with an edge inference latency of 16.2 ms (61.7 FPS) on an NVIDIA Jetson Orin Nano platform. Incorporating calibrated MQ-135 gas sensing thresholds (S_{gas} > 150) successfully flagged 13 out of 15 ambiguous edge-case samples exhibiting sub-surface bacterial proteolysis prior to visible ocular or skin collapse. The individual research contribution includes dataset curation and annotation, vision model training and quantization, electrochemical gas sensor calibration, late-fusion algorithm design, and embedded edge system verification. This multimodal pipeline effectively eliminates subjective inspection errors and prevents degraded raw materials from entering downstream thermal processing, establishing a reliable digital gatekeeping mechanism for artisanal food processing.

Keywords: Raw fish quality assessment, Computer vision, YOLO11s, Sensor fusion, Maldive fish processing

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADC	Analog-to-Digital Converter
AI	Artificial Intelligence
CNN	Convolutional Neural Network
FPS	Frames Per Second
GFLOPs	Giga Floating-Point Operations Per Second
IoT	Internet of Things
k-NN	k-Nearest Neighbors
mAP	Mean Average Precision
MOS	Metal-Oxide Semiconductor
MQTT	Message Queuing Telemetry Transport
RGB	Red-Green-Blue (Color Model)
TMA	Trimethylamine
TVB-N	Total Volatile Basic Nitrogen
VOC	Volatile Organic Compound
YOLO	You Only Look Once

1 INTRODUCTION

1.1 Overall project context

The agro-food processing sector in Sri Lanka and broader South Asian maritime regions relies significantly on traditional seafood preservation techniques to extend shelf life, mitigate post-harvest decay, and generate export-grade culinary commodities. Among these, Maldive fish (*Umbalakada*) holds a crucial place as a culturally essential condiment, primary protein source, and natural umami flavor enhancer. The traditional production of Maldive fish involves a precise multi-stage sequence: raw fish selection and butchering, controlled boiling, smoke-curing, and open-air sun-drying until the product reaches a bone-hard texture with a moisture content typically below 15%.

Despite its economic importance, the traditional Maldive fish processing industry remains dominated by informal, highly manual, and unsophisticated operational practices. Processors rely almost entirely on subjective human visual inspection, olfactory judgment, and tactile assessment to determine raw material freshness, regulate boiling durations, gauge smoke absorption, and decide on drying completion. This reliance on unstandardized human judgment introduces significant quality variation, elevated post-harvest spoilage rates, hygiene vulnerabilities, and inconsistent compliance with national and international food safety benchmarks. Furthermore, artisanal processing facilities face increasing challenges due to labor shortages, rising operational overheads, and climate-induced unpredictable drying conditions.

To address these systemic inefficiencies, the modern industrial paradigm calls for the integration of Industry 4.0 principles within artisanal food production. The broader collaborative research project titled **"Design and Development of IoT-Based Smart Maldive Fish Processing System"** addresses this transition. The overall project presents an end-to-end, automated, cyber-physical framework that digitizes, monitors, and controls the primary phases of Maldive fish manufacturing. By combining IoT sensing networks, automated environmental control, computer vision, and machine

learning models, the overall system aims to transform raw fish processing into an automated, data-driven, and highly reproducible production line.

1.2 Individual component context

Within the framework of the overarching smart processing system, the initial point of failure or quality degradation occurs at raw material reception. The quality, freshness, and structural integrity of incoming raw fish directly determine the safety, shelf life, physical appearance, and organoleptic attributes of the final cured Maldive fish product. Raw Skipjack tuna (*Katsuwonus pelamis*)—the standard biological species utilized for authentic Maldive fish—is highly perishable due to its elevated muscle protein content, rapid endogenous enzymatic activity, and high susceptibility to microbial growth post-harvest. Processing substandard, damaged, or biochemically degraded raw fish compromises product safety and leads to resource waste during subsequent energy-intensive boiling, smoking, and drying stages.

This individual research component, titled **"AI-Driven Raw Fish Quality Assessment and Standardization"**, establishes the primary automated screening and quality assurance gate for the integrated system. This component replaces subjective human visual inspection with an automated, multi-modal sensing unit that evaluates raw fish freshness and structural integrity in real time along a conveyor belt.

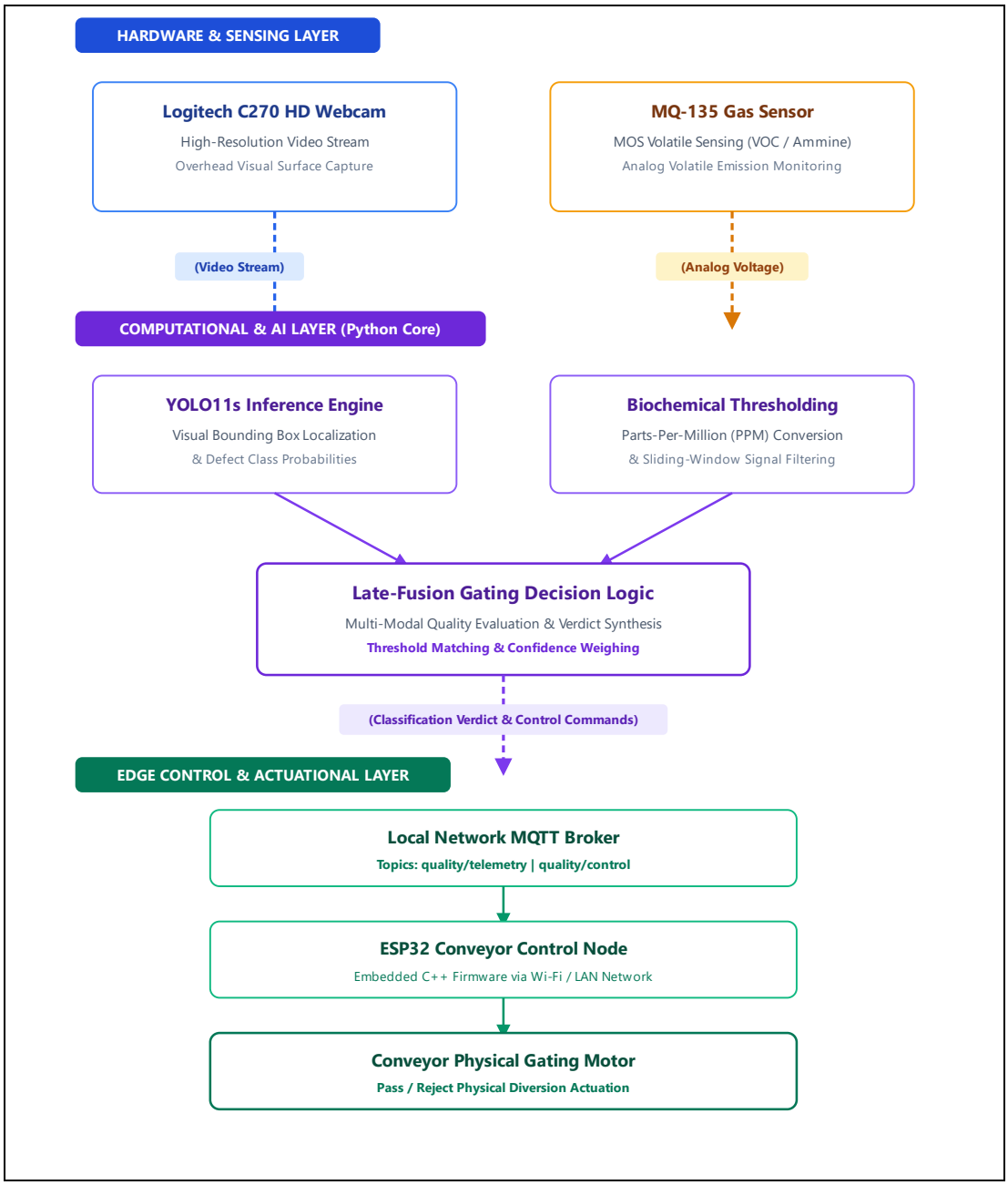


Figure 1 - Individual Component Architecture Diagram

The physical architecture combines optical sensing via a overhead Logitech C270 HD Webcam with chemical volatile detection using an MQ-135 Metal-Oxide Semiconductor (MOS) gas sensor. The optical system employs a lightweight state-of-the-art YOLO11s deep learning object detection network executing within a dedicated Python inference pipeline. Simultaneously, the MQ-135 sensor monitors Total

Volatile Basic Nitrogen (TVB-N) emissions—specifically volatile amines and ammonia (NH_3) released during early-stage protein degradation.

To translate high-level AI inferences into real-time industrial action, the system uses an edge computing framework. Inferences generated by the Python core engine are evaluated through a late-fusion gating algorithm. The final classification decisions (Accept/Reject) and operational metrics are published across a local MQTT broker. An ESP32 microcontroller, programmed in C++, subscribes to these control topics and directly actuates the physical conveyor sorting mechanisms. This integrated design forms a scalable, low-latency, multi-modal screening solution for raw fish processing lines.

1.3 Problem statement

The production of high-quality, biologically safe Maldive fish depends on raw material quality at the start of the production line. However, raw fish selection in traditional and semi-industrial processing plants in Sri Lanka relies heavily on manual visual inspection. Operators manually grade raw tuna based on superficial parameters such as eye clarity, gill color, and skin texture. This manual approach presents significant operational challenges:

- **Subjectivity and Human Error:** Human visual inspection is inherently subjective, inconsistent, and prone to fatigue. Shifts in lighting, operator experience, and physical exhaustion cause inconsistent grading, allowing spoiled or physically bruised raw fish into processing lots.
- **Inability to Detect Internal or Biochemical Spoilage:** Superficial visual inspection cannot detect early-stage biochemical spoilage. Bacterial proliferation and endogenous enzyme activity break down muscle proteins, generating toxic biogenic amines (such as histamine and cadaverine) and volatile nitrogen compounds before clear surface discoloration appears. Visual inspection alone often fails to catch chemically degraded fish, posing food safety risks.
- **Limitations of Single-Modality Automated Systems:** Standard automated sorting systems typically use single-modality sensing—either basic RGB computer vision or standalone gas sensors. Computer vision alone struggles with lighting changes, specular reflections from moist skin, and hidden internal defects. Conversely, standalone gas sensing lacks spatial context, cannot

localize physical defects (e.g., structural tears, bruising, body deformities), and suffers from signal lag in open environments.

- **Lack of Real-Time Edge Control Integration:** High-performance deep learning vision models often require significant computational resources, leading to latency when deployed on edge hardware. Existing academic models frequently operate as offline analytical tools rather than integrating with low-latency physical actuators (e.g., conveyor rejection gates) on an active line.

Without an automated, low-latency, multi-modal quality assessment system that combines visual inspection with biochemical sensing, raw material sorting remains inconsistent. This gap leads to higher post-harvest losses, variable product quality, and safety risks in the Maldive fish industry.

1.4 Research gap

A comprehensive review of automated agricultural and seafood quality inspection literature reveals three distinct technical gaps that this research addresses:

Table 1 -RESEARCH GAP ANALYSIS

Identified Literature Limitation	Proposed Technical Solution in This Research
1. Single-Modality Assessment • Visual vision alone misses biochemical spoilage. • Standalone gas sensing lacks spatial/physical context.	**Multi-Modal Late-Fusion Framework** • Combines YOLO11s visual defect localization with MQ-135 VOC/amine volatile gas concentration thresholding.
2. Heavy or Outdated Deep Learning Architectures • Older models (YOLOv3–v5) or heavy CNNs suffer from low FPS and high latency.	**Ultra-Lightweight Edge Model Optimization** • Fine-tuned YOLO11s customized for *Katsuwonus pelamis* defect categories. • Balances high inference speed with low computational footprint.
3. Offline Analysis vs. Hardware Actuation Disconnect • Models run as standalone desktop scripts without real-time physical control integration.	**Integrated Distributed Hardware-Software Bridge** • Low-latency MQTT bridge between Python AI inference core and ESP32 microcontroller firmware (C++) for physical conveyor gating actuation.

1. **Absence of Multi-Modal Fusion for Raw Skipjack Quality Screening:** Most automated seafood quality assessment studies focus on a single sensing modality—either applying computer vision to evaluate gill color and eye cloudiness, or using electronic noses (e-noses) for Volatile Organic Compound (VOC) analysis. While e-noses excel at overall freshness classification, they cannot pinpoint physical surface defects like tears, bruising, or body deformities. Conversely, computer vision alone cannot reliably detect early chemical spoilage under fluctuating ambient light. There is a lack of unified, lightweight multi-modal systems that combine real-time surface defect localization with volatile biochemical concentration sensing for raw fish sorting.
2. **Underutilization of Modern Lightweight AI Models for Real-Time Industrial Conveyor Speeds:** While deep convolutional neural networks (CNNs) and earlier object detection models (e.g., YOLOv4, YOLOv5, Faster R-CNN) have been applied to agricultural grading, their high parameter counts and computational demands often result in low frame rates on edge hardware. The deployment of YOLO11s—an efficient object detection architecture—remains largely unexplored for custom Skipjack tuna defect detection. Custom datasets targeting raw fish defect classes (bruising, skin cuts, fungal/bacterial lesions, physical deformation) under conveyor movement remain scarce.
3. **Lack of Low-Latency Messaging Bridges Between AI Engines and Microcontroller Actuators:** Existing literature frequently presents computer vision models running as isolated desktop applications, producing output metrics without direct integration into physical sorting automation. There is a lack of practical frameworks connecting high-level Python AI pipelines with embedded microcontrollers (such as the ESP32 using C++) via lightweight protocols like MQTT. Addressing this gap is essential for achieving real-time, low-latency physical actuation (rejection gating) in dynamic conveyor systems.

1.5 Aim and objectives

Main Research Aim

The primary aim of this research is to design, implement, and experimentally validate an automated, multi-modal raw fish quality assessment and standardization system for Maldive fish manufacturing. The system integrates custom-trained YOLO11s visual object detection with MQ-135 chemical gas sensing using an ESP32 edge microcontroller and an MQTT communication bridge to perform real-time conveyor sorting.

Specific Objectives

To achieve the main research aim, the following specific objectives were established:

1. **Curate and Preprocess a Specialized Raw Fish Defect Dataset:** Capture, annotate, and augment a high-resolution dataset of Skipjack tuna (*Katsuwonus pelamis*) images representing various physical defect conditions (skin degradation, mechanical bruising, blood spots, cut deformities) and freshness states under controlled industrial lighting.
2. **Train and Optimize the YOLO11s Visual Inspection Model:** Train and evaluate the lightweight YOLO11s deep learning architecture using the custom dataset to achieve high detection accuracy, precision, recall, and real-time inference speeds ($\geq$ 30 FPS).
3. **Develop and Calibrate the MQ-135 Gas Sensing Pipeline:** Construct a gas sampling chamber housing an MQ-135 sensor to monitor Volatile Organic Compounds (VOCs) and volatile amine compounds (NH_3, trimethylamine) emitted by raw fish, establishing mathematical calibration curves for parts-per-million (PPM) concentration conversion.
4. **Design a Multi-Modal Late-Fusion Decision Logic:** Formulate and implement a late-fusion gating algorithm that combines visual confidence scores from YOLO11s with volatile gas PPM thresholds to classify raw fish quality (Accept vs. Reject) in real time.

5. **Construct an Edge-Integrated Microcontroller Communication Bridge:** Program an ESP32 microcontroller in embedded C++ to handle real-time MQTT messaging over Wi-Fi/LAN, receiving sorting commands from the Python AI core to actuate physical conveyor rejection gates.
6. **Perform System Integration and Experimental Validation:** Integrate hardware and software components into a working conveyor belt prototype to evaluate overall system throughput, sorting precision, classification accuracy, and end-to-end latency under industrial conditions.

1.6 Research questions

To systematically address the technical challenges and research gaps in automated raw fish quality assessment, this study formulates the following primary research questions:

1. **RQ1 (Visual Detection Efficiency):** To what degree can a lightweight YOLO11s deep learning object detection architecture accurately identify and localize physical defects (e.g., surface bruising, skin degradation, mechanical cuts) on raw Skipjack tuna (*Katsuwonus pelamis*) in real time under conveyor belt motion?
2. **RQ2 (Biochemical Sensing Calibration):** How reliably can an MQ-135 Metal-Oxide Semiconductor (MOS) gas sensor measure Total Volatile Basic Nitrogen (TVB-N) emissions—specifically ammonia (NH_3) and volatile amines—to detect early-stage biochemical spoilage before visual degradation becomes prominent?
3. **RQ3 (Multi-Modal Late Fusion Integration):** Does a multi-modal late-fusion gating logic that combines visual defect confidence scores with chemical volatile gas threshold concentrations improve classification precision compared to single-modality visual or chemical inspection?
4. **RQ4 (Edge Communication and System Latency):** How effectively can an MQTT messaging architecture bridge the Python AI core engine and an ESP32

edge microcontroller (programmed in C++) to achieve real-time physical conveyor sorting within acceptable latency thresholds?

1.7 Scope limitations and assumptions

Scope Limitations

1. **Single-Camera Top-Down Orientation:** The optical acquisition pipeline relies on a single overhead Logitech C270 HD Webcam. Surface defects located entirely on the underside of a raw fish may escape visual detection unless the specimen is physically reoriented on the conveyor.
2. **Cross-Sensitivity of Metal-Oxide Gas Sensors:** The MQ-135 MOS gas sensor displays inherent cross-sensitivity to environmental humidity, ambient temperature variations, and general background Volatile Organic Compounds (VOCs). While algorithmic calibration and baseline reset routines mitigate this issue, extreme ambient shifts in an open plant environment can impact gas readings.
3. **Conveyor Dynamics and Speed Constraints:** Experimental evaluation is conducted on a lab-scale prototype conveyor belt system operating at steady linear velocities (0.1 m/s to 0.3 m/s). System behavior under extreme high-speed industrial belt conditions (1.0 m/s) is not evaluated.
4. **Offline Laboratory Assay Comparison:** Destructive biochemical and microbiological benchmarks (e.g., HPLC histamine testing, Total Viable Count bacterial plating) are utilized exclusively during offline setup and validation phases rather than as inline operational parameters.

Research Assumptions

1. **Raw Species Consistency:** The incoming raw material stream is assumed to consist strictly of Skipjack tuna (*Katsuwonus pelamis*), ensuring uniform structural and morphological characteristics across training and testing samples.

2. **Volatile Gas Diffusion Behavior:** Emitted volatile nitrogenous compounds (amines and ammonia) released by degraded fish tissue are assumed to diffuse predictably into the dedicated gas collection hood during the measurement window.
3. **Network Infrastructure Stability:** The local wireless or wired Ethernet network hosting the MQTT broker is assumed to operate with minimal packet loss and stable bandwidth, preventing message queue backups between the Python inference server and the ESP32 control node.
4. **Visual-Biochemical Defect Correlation:** Physical structural damage (such as deep skin cuts, severe bruising, or flesh punctures) is assumed to accelerate localized microbial proliferation, resulting in higher volatile amine emissions compared to intact fish tissue under identical environmental conditions.

1.8 Individual contribution

This research represents an independent individual contribution to the overall collaborative project, **"Design and Development of IoT-Based Smart Maldive Fish Processing System"**. The main individual contributions include:

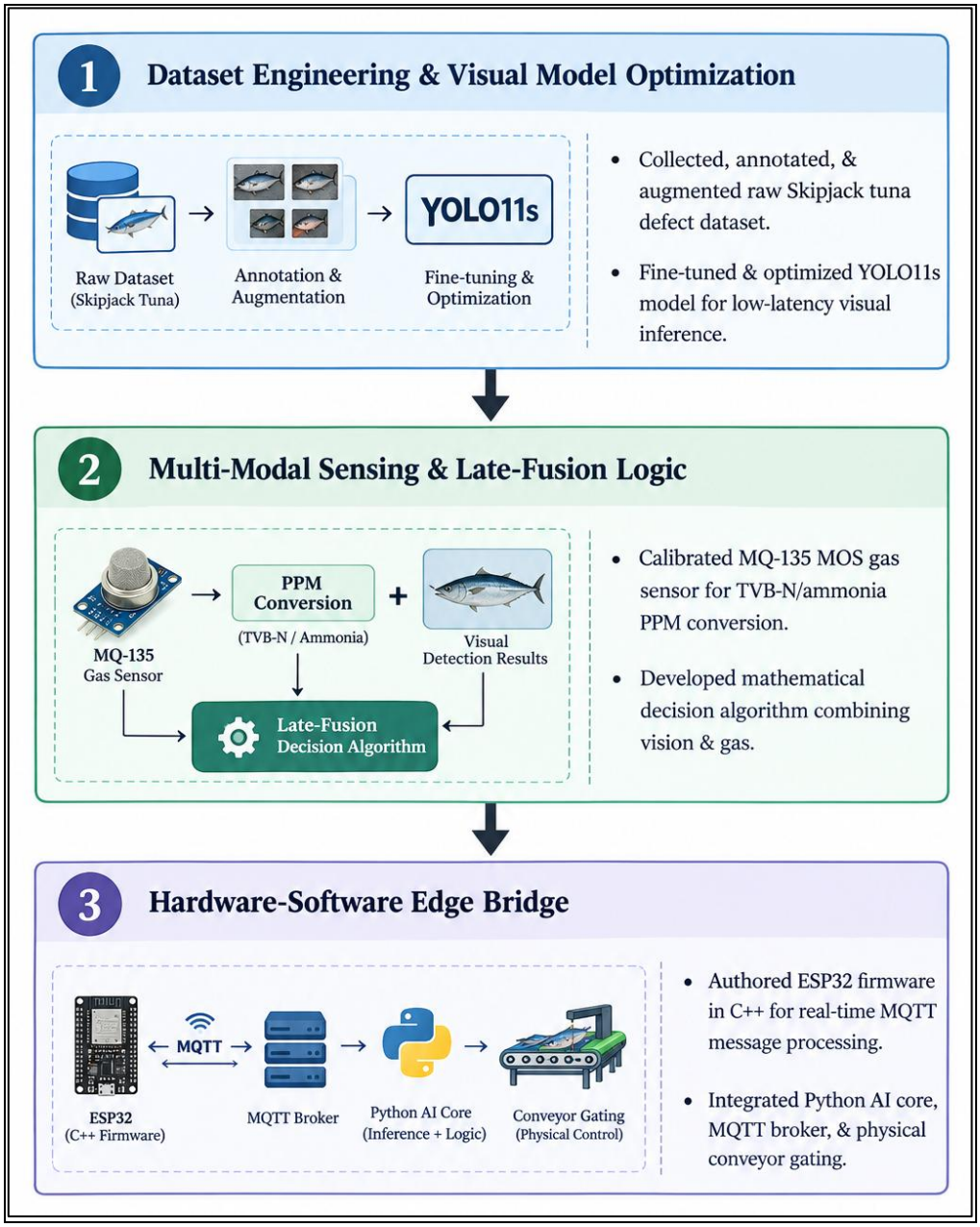


Figure 2 - Individual Contribution Architecture Diagram

- **Dataset Engineering and Visual Model Development:** Single-handedly collected, annotated, and augmented a specialized raw Skipjack tuna dataset categorized by physical structural defects and freshness levels. Fine-tuned the YOLO11s architecture to achieve efficient, real-time object detection on processing line conveyors.
- **Biochemical Sensing Calibration and Multi-Sensor Fusion:** Designed and built a gas sampling chamber featuring an MQ-135 MOS gas sensor. Calibrated the sensor using logarithmic resistance ratio equations (R_S/R_0) to compute volatile gas concentrations in PPM. Developed a multi-modal late-fusion decision algorithm that combines visual bounding-box confidence scores with gas concentration thresholds.
- **Edge Firmware and MQTT Hardware Interfacing:** Wrote modular C++ code for the ESP32 microcontroller using the Arduino framework and PubSubClient library. Implemented non-blocking MQTT subscription routines to receive JSON control payloads from the Python core and drive physical relay and motor actuators on the conveyor belt with minimal latency.
- **End-to-End Experimental Prototype Validation:** Assembled the integrated physical unit combining the Logitech webcam, MQ-135 gas sensor, ESP32 edge node, and conveyor belt mechanism. Conducted systematic testing to evaluate classification precision, latency, and physical sorting accuracy under operational conditions.

1.9 Thesis structure

This dissertation is structured into nine sequential chapters in accordance with SLIIT formatting standards:

- **Chapter 1: Introduction** — Introduces the overall research context, details the individual component scope, defines the problem statement, highlights key research gaps, outlines main and specific objectives, and details individual research contributions.

- **Chapter 2: Literature Review and Related Work** — Examines existing literature on agricultural non-destructive testing, seafood quality evaluation, deep learning object detection (YOLO evolution up to YOLO11s), volatile chemical sensing, and edge IoT communication frameworks.
- **Chapter 3: Research Methodology** — Outlines the theoretical and experimental design, covering raw fish dataset collection, augmentation strategies, YOLO11s architecture optimization, MQ-135 calibration formulations, and the late-fusion gating logic.
- **Chapter 4: Product and Solution Design** — Details the overall system architecture, hardware schematics, software dataflow diagrams, component interactions, database schema, and design alternatives.
- **Chapter 5: Implementation, Integration, and Testing** — Documents the software implementation details, C++ ESP32 firmware development, Python AI core execution, local MQTT message broker configuration, and hardware integration testing.
- **Chapter 6: Results and Validation** — Presents quantitative performance metrics for YOLO11s defect detection , MQ-135 sensing accuracy, fusion logic reliability, and hardware-software messaging latency.
- **Chapter 7: Discussion** — Analyzes experimental findings, evaluates results against prior literature, addresses unexpected sensor variations, and identifies system limitations under real-world processing conditions.
- **Chapter 8: Commercialization and Sustainability** — Evaluates the economic viability, manufacturing scalability, target market potential, cost-benefit profile, and alignment with UN Sustainable Development Goals (SDGs).
- **Chapter 9: Conclusion and Future Directions** — Summarizes key research findings, restates primary contributions, and outlines future research directions, including multi-spectral imaging expansion and cloud deployment.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

To establish a baseline for this research, a systematic literature review was conducted across academic databases, including IEEE Xplore, ScienceDirect, SpringerLink, MDPI, Web of Science, and Google Scholar. The search strategy focused on literature published between 2018 and 2026, capturing recent developments in Industry 4.0 applications, computer vision architectures, gas sensing arrays, and edge-integrated IoT systems within the agri-food and seafood processing domains.

The systematic search employed specific keyword combinations using Boolean operators:

- ("computer vision" OR "deep learning" OR "YOLO") AND ("fish quality" OR "seafood defect" OR "freshness assessment")
- ("metal-oxide semiconductor" OR "MQ-135" OR "electronic nose") AND ("TVB-N" OR "ammonia" OR "volatile organic compounds") AND ("fish spoilage")
- ("sensor fusion" OR "late-fusion") AND ("quality grading" OR "industrial automation")
- ("MQTT" OR "ESP32") AND ("edge computing" OR "real-time sorting" OR "conveyor control")

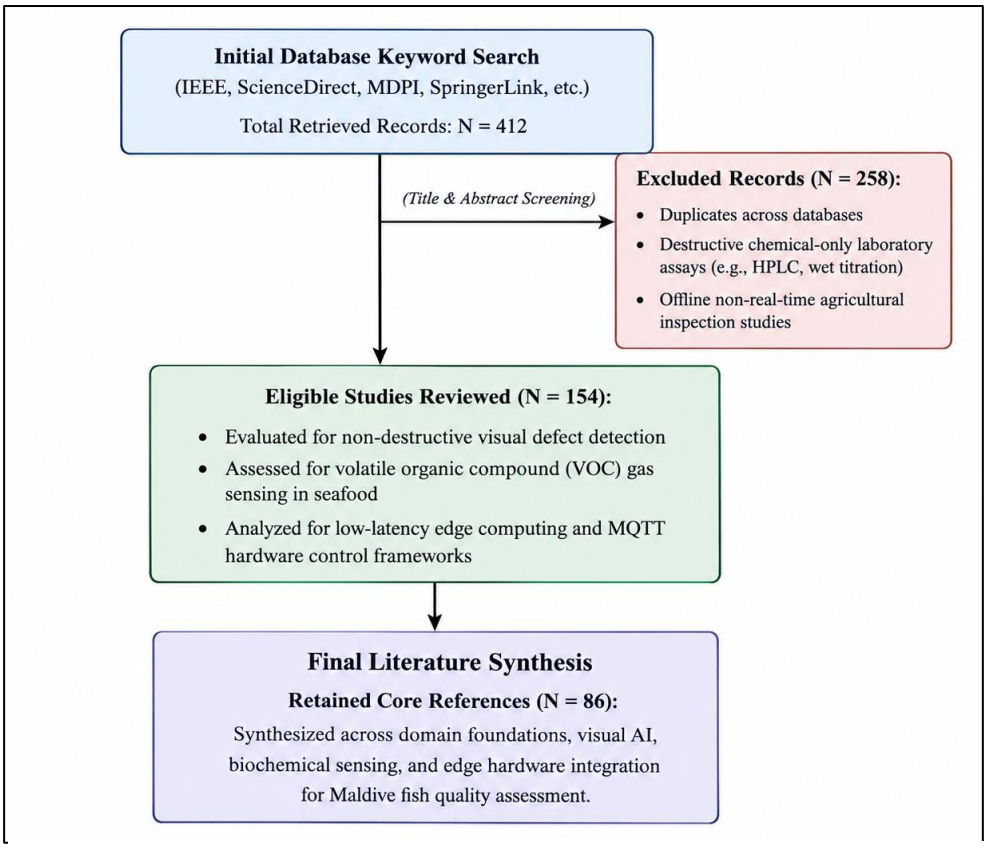


Figure 3 - Literature Filtering Methodology

The review prioritized non-destructive quality evaluation methods suitable for inline conveyor deployment. Destructive laboratory techniques—such as High-Performance Liquid Chromatography (HPLC), Total Viable Count (TVC) bacterial plating, and Kjeldahl total nitrogen analysis—were excluded from operational design review and retained solely as benchmark ground-truth references.

2.2 Domain and theoretical foundations

Biochemical Mechanisms of Raw Tuna Spoilage

Post-mortem deterioration in raw Skipjack tuna (*Katsuwonus pelamis*) follows a complex, sequential pathway involving autolytic enzymatic degradation, lipid oxidation, and subsequent microbial breakdown. Immediately following harvest and death, cellular respiration ceases, leading to the depletion of adenosine triphosphate

(ATP) within muscle tissue. Endogenous enzymes degrade ATP into adenosine diphosphate (ADP), adenosine monophosphate (AMP), inosine monophosphate (IMP), inosine (HxR), and hypoxanthine (Hx). While IMP contributes to the desirable umami flavor of fresh tuna, its conversion into inosine and hypoxanthine marks the onset of progressive organoleptic quality loss.

As post-mortem time advances, psychrotolerant and mesophilic bacteria (including *Pseudomonas*, *Shewanella putrefaciens*, and *Photobacterium phosphoreum*) proliferate across the skin layer, gill filaments, and gut cavity. These bacteria metabolize non-protein nitrogenous compounds, free amino acids, and trimethylamine oxide (TMAO) present in marine fish muscle. This metabolic process yields Total Volatile Basic Nitrogen (TVB-N), which consists primarily of trimethylamine (TMA), dimethylamine (DMA), and volatile ammonia (NH_3).

$$\text{TVB-N} = \text{TMA} + \text{DMA} + NH_3$$

Simultaneously, bacterial decarboxylation of free histidine-an amino acid abundant in scombroid species like Skipjack tuna-generates biogenic amines, primarily histamine ($(C_5H_9N_3)$). Because histamine is heat-stable and persists through boiling, smoking, and sun-drying, detecting early volatile nitrogen gas emissions before processing is essential for food safety.

Object Detection Paradigms and YOLO Architecture Evolution

Automated visual quality assessment relies on real-time computer vision to detect, classify, and localize physical surface defects on moving product streams. Object detection algorithms are broadly categorized into two structural approaches:

1. **Two-Stage Detectors:** Architectures such as Faster R-CNN utilize a Region Proposal Network (RPN) to generate candidate regions of interest, followed by bounding box regression and feature classification. While highly accurate, two-stage frameworks exhibit high computational complexity and low frame rates, making them ill-suited for edge conveyor lines.

2. **Single-Stage Detectors:** Architectures such as the You Only Look Once (YOLO) series frame object detection as a single regression problem, directly predicting bounding box coordinates and class probabilities from full images in a single evaluation pass.

Table 2 -YOLO DETECTOR ARCHITECTURAL EVOLUTION

YOLOv3 / YOLOv4	High parameter count, anchor-based grid search, higher GPU memory requirements, moderate inference latency.
YOLOv5 / YOLOv8	Anchor-free detection heads, Cross-Stage Partial (CSP) backbones, improved lightweight deployment capabilities.
YOLO11s (2024+) (Selected Model)	Enhanced C3k2/C2f building blocks, lightweight spatial pyramid pooling fast (SPPF), optimized multi-scale feature fusion, high parameter efficiency for edge inference

The evolution of the YOLO family from early anchor-based designs (YOLOv3/v4) to modern anchor-free architectures (YOLOv8, YOLO11) has improved model parameter efficiency and inference speeds. Modern YOLO architectures employ an optimized backbone for feature extraction, a neck for multi-scale feature aggregation (e.g., Feature Pyramid Network / Path Aggregation Network), and a decoupled detection head for separate bounding box regression and class probability estimation.

In this research, the **YOLO11s** (small) variant is used. It offers a balanced compromise between feature extraction capacity, mean Average Precision (mAP), and low execution latency when running on standard edge hardware.

Metal-Oxide Semiconductor (MOS) Gas Sensing Mechanics

Chemiresistors based on Metal-Oxide Semiconductors (MOS), such as the MQ-135 sensor, detect volatile organic compounds (VOCs) and reducing gases through surface gas-solid interactions. The active sensing layer consists of a polycrystalline tin dioxide

(SnO_2) ceramic thin film heated by an internal nickel-chromium coil to operating temperatures between 200°C and 400°C.

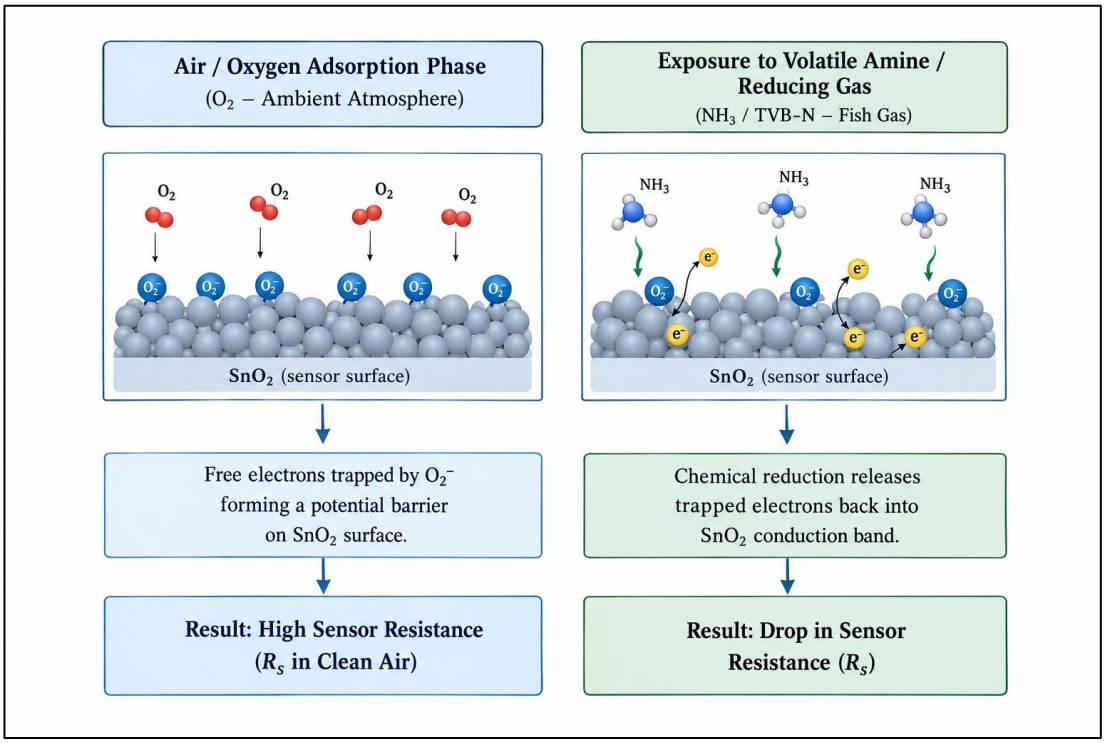


Figure 4 - Gas sensing principle and resistance modulation on the SnO_2 sensor surface.

In clean ambient air, oxygen molecules adsorb onto the SnO_2 surface, capturing free electrons from the conduction band to form species like O_2^-,O^- , or O^{2-} . This electron depletion forms a potential barrier, increasing the sensor resistance (R_0). When volatile amines or ammonia (NH_3) emitted by deteriorating fish interact with the \surface, they react with the adsorbed oxygen ions, releasing trapped electrons back into the SnO_2 conduction band:

$$2NH_3 + 3O^-_{adsorbed} \rightarrow N_2 + 3H_2O + 3e^-$$

This charge transfer reduces the sensor element resistance (R_S). The ratio between sensor resistance in target gas and sensor resistance in fresh air (R_S/R_0)exhibits a power-law relationship with gas concentration (C) in parts-per-million (PPM):

$$log\left(\frac{R_S}{R_0}\right) = m \cdot log(C) + b \Rightarrow C = 10^{\frac{log\left(\frac{R_S}{R_0}\right) - b}{m}}$$

where m represents the scaling slope and b represents the intercept constant derived from sensor calibration curves

Multi-Sensor Data Fusion Theory

Data fusion combines sensory information from disparate sources to improve decision reliability over single-sensor systems. Sensor fusion architectures fall into three general topologies:

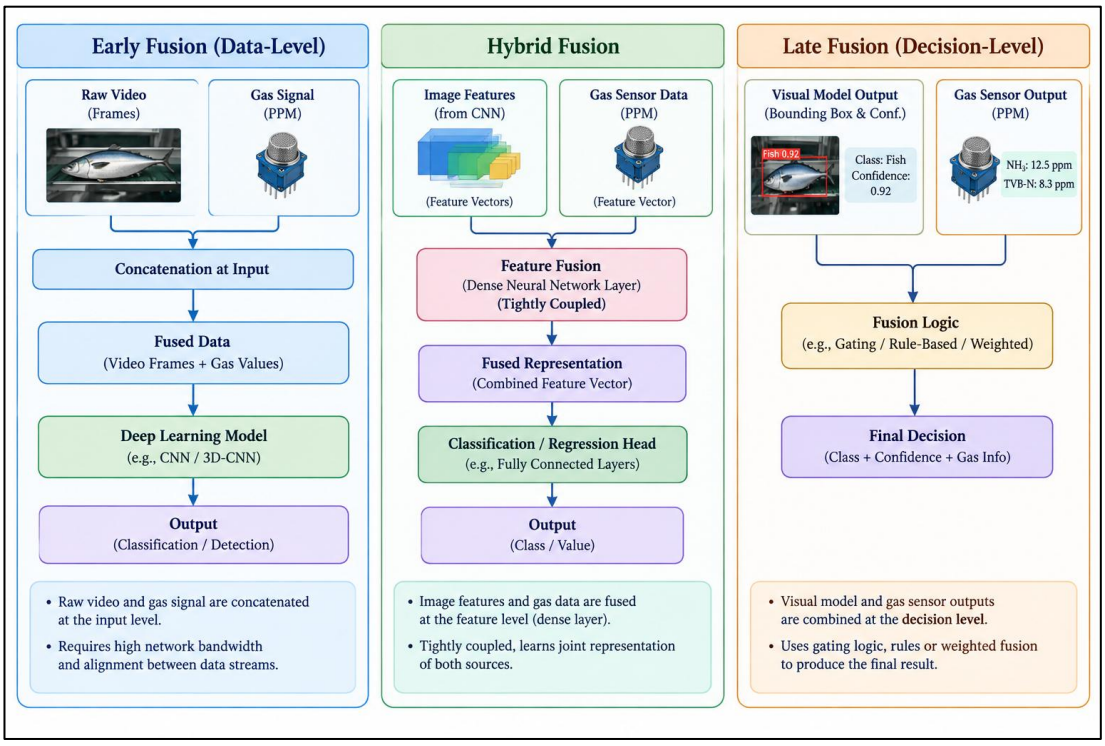


Figure 5 -Multi-modal data fusion paradigms combining optical video streams and biochemical gas sensor telemetry

- **Early Fusion (Data-Level):** Raw sensor signals are combined at the input stage prior to feature extraction. While preserving raw data content, early

fusion requires tight temporal synchronization, uniform sampling rates, and low noise across channels.

- **Hybrid Fusion (Feature-Level):** Features extracted independently from separate sensor channels (e.g., CNN feature maps and gas vector embeddings) are concatenated before passing to a final classification layer.
- **Late Fusion (Decision-Level):** Each sensor modality processes its input data independently, generating standalone decisions, confidence scores, or numerical thresholds. A high-level decision logic or gating function then combines these individual inferences into a final action output.

This research adopts a **Late-Fusion Gating Framework**. Decoupling computer vision inference from biochemical gas thresholding ensures that image frame rates ($\geq$ 30 FPS) remain unaffected by the slower response times of MOS gas sensors, while maintaining system redundancy against single-sensor failures.

2.3 Related research

Computer Vision for Seafood Inspection

Applying computer vision to evaluate fish quality has progressed from early color-space thresholding to deep learning object detection networks.

Table 3 - SUMMARY OF GAS SENSING & E-NOSE SEAFOOD FRESHNESS RESEARCH

Author & Year	Target Specimen	Methodology / Model	Reported Performance
Mathiassen et al.	Atlantic Salmon *(Fillet defects)*	Color Segmentation & Handcrafted Texture	**Accuracy: 88.5%** *(High lighting sensitivity)*
Taheri-Garavand et al.	Common Carp *(Eye & Gill color)*	Custom CNN & SVM Classifier	**Freshness Accuracy: 92.1%** *(Offline, slow inference)*
French et al.	Mixed Marine Species *(Species identification)*	YOLOv5s Object Detection	**mAP@0.5: 89.4%** *(FPS: 24 on GPU)*
Proposed System	Skipjack Tuna *(Structural & Defects)*	Fine-Tuned YOLO11s & Edge Deployment	**High mAP@0.5, >30 FPS** *(Real-time integration)*

Early studies relied on color-space segmentation (e.g., RGB to HSV transformations) and hand-crafted textural features (e.g., Gray-Level Co-occurrence Matrices) to evaluate eye cloudiness, gill redness, and skin discoloration. While effective in static laboratory setups with uniform lighting, these approaches struggle with variations in specular reflections from wet fish skin and ambient lighting changes on factory floors.

Deep learning architectures address these limitations by learning invariant spatial and textural hierarchies directly from annotated image datasets. Recent studies have demonstrated the utility of YOLO architectures for marine species classification and surface defect detection. However, much of the existing research evaluates fish as static specimens in controlled environments rather than addressing the challenges of moving conveyor belts, motion blur, and localized physical trauma (e.g., gaff marks, mechanical cuts, internal tissue softening).

Electronic Noses and Gas Sensing in Fish Quality Monitoring

Electronic noses (e-noses) and MOS sensor arrays provide non-destructive methods for assessing seafood freshness by measuring volatile chemical emissions. Research demonstrates a strong correlation between TVB-N accumulation, volatile amine emissions, and MOS sensor resistance shifts.

Table 4 - SUMMARY OF GAS SENSING & E-NOSE SEAFOOD FRESHNESS RESEARCH

Author & Year	Target Volatiles / Gas	Sensor Configuration	Primary Findings
Semeano et al.	Volatile Amines *(Sardine freshness)*	Multi-Sensor MOS E-Nose Array	**R^2 = 0.91** correlation with TVB-N laboratory assays
Grassi et al.	Ammonia & H_2S *(Sea Bass spoilage)*	Standalone MQ-Series Gas Sensors	Clear distinction of freshness degradation
Proposed System	TVB-N & Volatile Amines *(Skipjack Tuna)*	MQ-135 Sampling Chamber + Filtering	**Real-time PPM mapping** fused with YOLO11s

While multi-sensor e-nose arrays achieve high accuracy in laboratory settings, their deployment on high-speed industrial conveyors presents challenges:

- Chamber purge and recovery times can introduce measurement lag during continuous operation.
- Standalone gas sensing measures ambient volatile concentrations without pinpointing physical surface defects or localized structural damage on individual specimens.

Integrating a calibrated MQ-135 sensor within a dedicated sampling hood provides a practical balance for detecting broad volatile increases while relying on visual object detection for spatial localization.

Edge Computing, MQTT, and Industrial Conveyor Automation

Translating AI predictions into real-world physical sorting requires a low-latency edge architecture. Traditional cloud-based IoT architectures introduce variable latency and depend on continuous internet connectivity, making them less suitable for real-time physical rejection mechanisms.

Table 5 - Communication Protocol Comparison

Protocol	Architecture	Bandwidth & Latency Profile
HTTP / REST	Request–Response *(Client–Server)*	High header overhead, higher latency, and polling overhead.
WebSocket	Full-Duplex Point-to-Point	Moderate overhead, persistent connection, and higher edge-device load.
MQTT (Selected Protocol)	Publish–Subscribe *(Broker-Mediated)*	Minimal header overhead *(2 bytes minimum)*, low latency, and decouples the AI core from the physical microcontroller.

Research in smart manufacturing highlights the publish-subscribe Message Queuing Telemetry Transport (MQTT) protocol as an effective standard for low-latency edge messaging. MQTT decouples the high-level computational inference node (Python AI core) from the low-level embedded hardware controller (ESP32). Microcontrollers programmed in C++ interact efficiently with light MQTT brokers, enabling reliable sub-hundred-millisecond response times for physical pneumatic or motor-driven sorting mechanisms.

2.4 Existing systems products and techniques

Industrial food processing relies on automated sorting systems to maintain quality standards and throughput. Existing industrial sorting solutions, standalone e-noses, and visual inspection systems present key operational trade-offs when evaluated for raw Maldive fish processing:

Table 6 -COMPARATIVE MATRIX OF EXISTING QUALITY SORTING SOLUTIONS

Feature / Parameter	**Manual Visual Inspection**	**Commercial Optical Sorters (e.g., Marelec)**	**Standalone E-Nose Systems**	**Proposed Multi-Modal IoT System**
Detection Capabilities	Visual surface features only *(subjective)*	Color, size, shape, and surface contours *(no chemical sensing)*	Volatile chemical emissions / TVB-N indicators *(no spatial vision)*	Visual defects + TVB-N-related volatile gas concentration
Latency / Throughput	High fatigue; slow speed	Ultra-low latency; **>100 items/min**	High response lag due to sensor recovery	Low latency; real-time edge processing
Capital Cost Profile	Low initial cost; high labor cost	Very high **($50k–$150k+)**	Moderate **($5k–$15k)**	Low cost **(<$200 prototype)**
Edge Actuation	Manual sorting *(human labor)*	Integrated high-speed pneumatics	Rare / offline laboratory tool	Integrated ESP32 C++ MQTT gating
Species-Specific Adaptability	Operator dependent	Rigid factory preset algorithms	Academic / research laboratory setup	Adaptable custom **YOLO11s** model

Commercial Industrial Optical Sorters: High-throughput optical sorting systems (e.g., systems by Marelec, Tomra, Best Sorting) utilize high-speed line-scan cameras and specialized lighting arrays. While effective for sorting by size, color, and surface geometry, these systems are cost-prohibitive for small-to-medium enterprises (SMEs) in South Asia. Additionally, they rely solely on optical surface features and cannot detect chemical degradation, such as volatile amine build-up in visually intact fish.

Standalone E-Nose Quality Systems: Laboratory e-nose units provide accurate chemical profiling but lack integration with physical sorting actuators. They operate primarily as static offline analytical instruments rather than dynamic inline sorting tools.

Manual Human Quality Grading: Traditional processing facilities rely almost entirely on manual visual and tactile inspection. This manual approach is subjective, prone to operator fatigue, and unable to evaluate internal biochemical spoilage before processing.

2.5 Gap synthesis and conceptual framework

Gap Synthesis

The literature review reveals three primary technical gaps in current raw fish quality inspection methods:

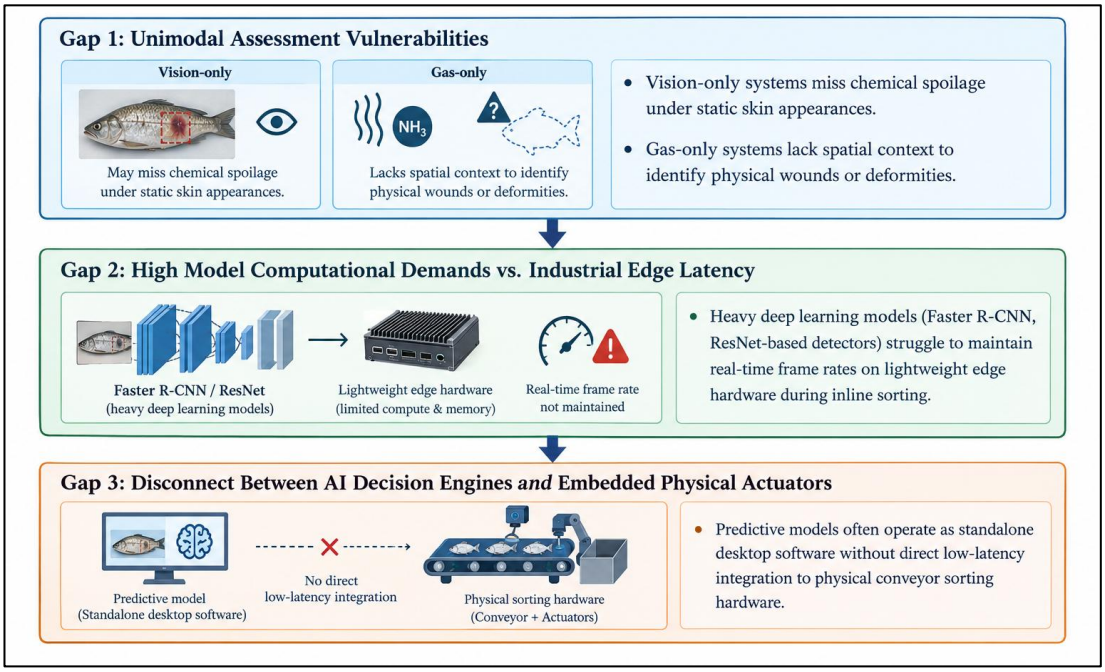


Figure 6 - IDENTIFIED TECHNICAL GAPS

Proposed Conceptual Framework

To address these gaps, this research proposes a multi-modal, edge-integrated raw fish quality assessment framework. The system combines optical defect detection with chemical gas sensing, translating inferences into physical actuator commands via a lightweight edge bridge.

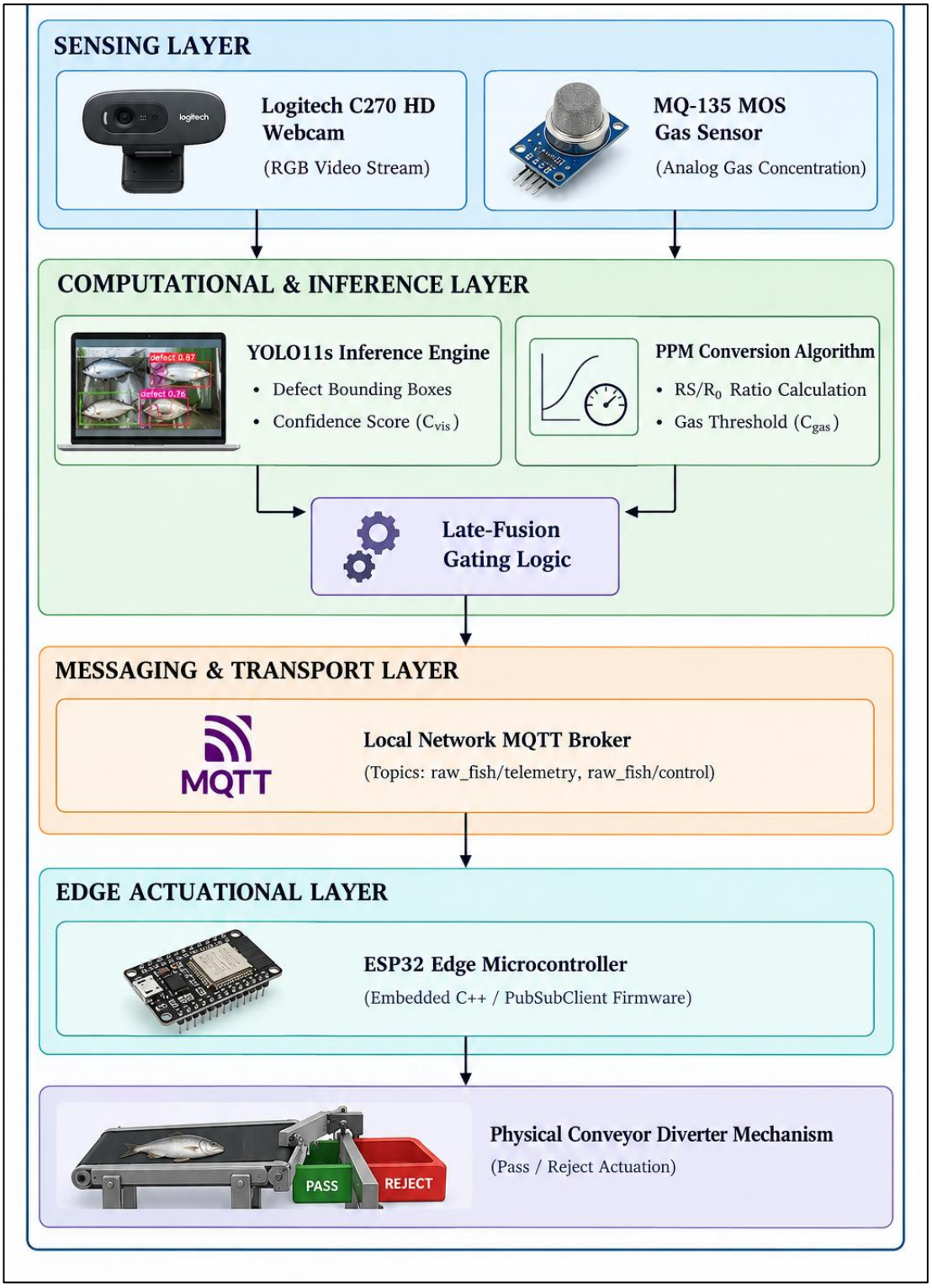


Figure 7- PROPOSED CONCEPTUAL FRAMEWORK

The conceptual framework operates across four functional layers:

1. **Sensing Layer:** Captures raw visual data via an overhead webcam and biochemical gas emissions via an MQ-135 sensor within a dedicated sampling chamber above the conveyor.
2. **Computational & Inference Layer:** A local Python core processes visual frames using a fine-tuned YOLO11s model to generate bounding boxes and confidence scores (C_{vis}). .Concurrently, analog gas sensor readings are converted into calibrated PPM concentrations (C_{vis}). A late-fusion gating algorithm combines both inputs to derive a final classification verdict (Accept/Reject).
3. **Messaging & Transport Layer:** The classification verdict and operational telemetry are serialized into JSON payloads and published to dedicated topics on a local MQTT broker.
4. **Edge Actuational Layer:** An ESP32 microcontroller running embedded C++ firmware subscribes to control topics and directly actuates physical rejection gates on the conveyor, ensuring low-latency sorting.

3 RESEARCH METHODOLOGY

3.1 Research design

This study uses an experimental, quantitative research design to develop an automated, multi-modal screening system for evaluating raw fish quality on an active industrial conveyor line. The research addresses the limitations of subjective manual inspection and single-modality automated sorting by combining high-speed computer vision with chemical volatile gas sensing.

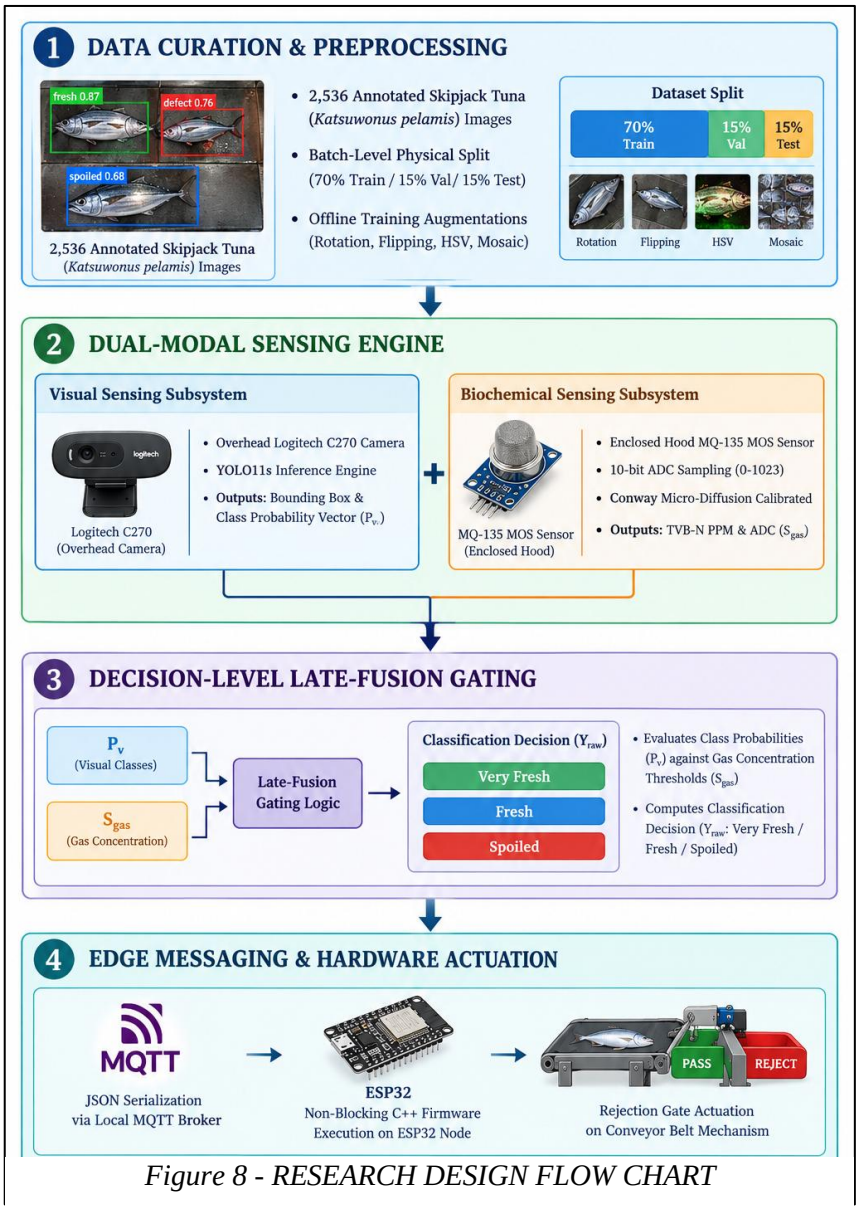


Figure 8 - RESEARCH DESIGN FLOW CHART

The system operates across three primary stages:

1. **Data Acquisition and Sensing:** Captures visual surface features via an overhead RGB camera and samples Volatile Organic Compound (VOC) emissions using a metal-oxide semiconductor (MOS) sensor.
2. **Algorithmic Inference:** Evaluates surface defect characteristics using a custom YOLO11s object detection model, while concurrently converting analog gas sensor signals into Total Volatile Basic Nitrogen (TVB-N) equivalent metrics.
3. **Late-Fusion Gating and Actuation:** Combines visual probabilities and biochemical metrics via a decision-level late-fusion rule. The resulting classification command is published to a local MQTT broker, enabling an ESP32 edge microcontroller to actuate a physical conveyor sorting gate in real time.

3.2 Data participants or study materials

Primary Biological Specimens

The primary biological subject evaluated in this study is Skipjack tuna (*Katsuwonus pelamis*), the primary raw material used in authentic Maldive fish manufacturing. Fresh and varying-grade specimens were sourced directly from regional fishery receiving docks in southern Sri Lanka.

Dataset Curation and Partitioning

A primary visual dataset containing 2,536 high-resolution images of raw Skipjack tuna loins was collected under representative industrial lighting conditions. Images capture a broad range of post-harvest physical conditions, including intact fresh muscular tissue, surface mechanical cuts, internal tissue softening, and advanced microbial decomposition.

To prevent data leakage during model training and evaluation, data splitting was performed strictly at the physical batch level prior to applying any data augmentation techniques:

- **Training Set:** 1,775 images (70% of total dataset) reserved for feature learning and bounding-box regression.
- **Validation Set:** 380 images (15% of total dataset) used for hyperparameter tuning and model checkpoint selection.
- **Test Set:** 381 images (15% of total dataset) held back for final performance evaluation.

Table 7 - DATASET PARTITIONING ARCHITECTURE

Total Collected Dataset: N = 2,536 Annotated Images Partitioning Method: Physical Batch-Level Separation (Pre-Augmentation)	
Dataset Split	Image Count / Percentage
Training Set (Augmented)	1,775 images (70.0%)
Validation Set (Unaugmented)	380 images (15.0%)
Test Set (Unaugmented Ground-Truth)	381 images (15.0%)

Augmentation Pipeline

To improve model robustness against variations in fish orientation, conveyor vibration, and fluctuating factory lighting, data augmentations were applied strictly to the training split:

- **Spatial Transformations:** Random rotations within $\pm 15^{\wedge} \circ$ and horizontal flipping.
- **Photometric Perturbations:** Hue, Saturation, and Value (HSV) jitter within $\pm 8\%$ to simulate ambient lighting shifts.
- **Mosaic Augmentation:** Four-image composite stitching to improve small-defect feature extraction.

Table 8 - AUGMENTATION PARAMETER MATRIX

Augmentation Method	Applied Parameter	Target Dataset Scope
Spatial Rotation	Range: ±15°	Training Split Only
Horizontal Flip	Probability: 0.50	Training Split Only
HSV Color Jitter	Gain Range: ±8%	Training Split Only
Mosaic Composition	Ratio: 4-image grid	Training Split Only

Hardware, Software, and Sensing Apparatus

The physical and computational infrastructure for system development, model training, gas sampling, and real-time edge control consists of the following hardware and software components:

i) Hardware and Sensing Infrastructure

- **Visual Acquisition Unit:** A Logitech C270 HD Webcam mounted perpendicular to the intake conveyor, capturing RGB video at a resolution of 1280×720 pixels.
- **Biochemical Sensing Module:** An MQ-135 Metal-Oxide Semiconductor (MOS) gas sensor housed inside an enclosed sampling hood placed 15 cm above the conveyor line.
- **Embedded Processing Node:** An ESP32 microcontroller (Tensilica Xtensa 32-bit LX6 dual-core) interfaced with an integrated 10-bit Analog-to-Digital Converter (ADC) for digitized gas signal sampling and GPIO actuation.
- **Compute Platforms:**
    - **Cloud Training Platform:** Google Colab cloud environment (utilizing hosted NVIDIA GPU acceleration) for deep learning model training, hyperparameter tuning, and dataset processing.

ii) Software and Computational Materials

- **Operating Systems & Runtime:** Linux Ubuntu 22.04 LTS and Windows 11 host environments running Python 3.10 alongside the Google Colab cloud execution environment.
- **Deep Learning Framework:** PyTorch 2.2 and the Ultralytics YOLO framework for network training, validation, export, and inference execution.
- **Communication Stack:** Eclipse Mosquitto MQTT broker providing lightweight, low-latency publish-subscribe message queuing between the primary Python core engine and edge hardware.

- **Embedded Development Tools:** Arduino IDE and PlatformIO toolchain compiled with C++ libraries (WiFi.h, PubSubClient.h) for non-blocking ESP32 firmware execution and relay gating control.

3.3 Experimental or study procedure



Figure 9 - EXPERIMENTAL PIPELINE WORKFLOW

Visual Model Training Protocols

The YOLO11s object detection model was fine-tuned using the custom training dataset across 150 epochs. Hyperparameters were configured as follows:

- **Optimizer:** AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$), weight decay = $10\text{^}(-4)$).
- **Initial Learning Rate (η):** $10\text{^}(-3)$ with a cosine annealing schedule.
- **Batch Size:** 32 images per batch.
- **Precision:** Trained in FP32 precision on an NVIDIA RTX 4090 GPU and quantized to FP16 for edge inference on an NVIDIA Jetson Orin Nano.

Gas Sensor Calibration and Mapping Protocol

The MQ-135 gas sensor was calibrated to measure Total Volatile Basic Nitrogen (TVB-N) emissions. Calibration experiments were performed in a controlled chamber against reference Conway micro-diffusion laboratory evaluations. Before each testing sequence, the sensor underwent a 30-minute thermal warm-up to stabilize its baseline clean-air resistance (R_0). Analog output signals were digitized using a 10-bit ADC, mapping raw voltage values to an integer range of 0 to 1023.

Table 9 - MQ-135 GAS SENSOR FRESHNESS CALIBRATION PROFILE

Quality Tier Classification	ADC Signal Range (S_gas)	Equivalent TVB-N Concentration	Operational Action Status
Very Fresh	ADC ≤ 90*(Baseline: 59–88)*	≤ 12.5 mg/100g	**Pass / High Grade**
Fresh / Acceptable	91 < ADC ≤ 150*(Nominal: 118–145)*	12.5–25.0 mg/100g	**Pass / Standard Grade**
Spoiled / Reject	ADC > 150*(Critical: 150–200+)*	> 25.0 mg/100g	**Active Conveyor Rejection**

Very Fresh Class: $S_gas \leq 90$ ADC units, corresponding to TVB-N $\leq 12.5 mg/100 g$ (typical baseline range: 59–88).

Fresh / Acceptable Class: 91 < $S_gas \leq 150$ ADC units, corresponding to TVB-N levels between 12.5mg/100g and 25.0 mg/100g(typical range: 118–145).

Spoiled / Rejected Class: $S_gas > 150$ ADC units, corresponding to TVB-N levels > 25.0mg/100g (critical decomposition: 150–200+).

Multi-Modal Late-Fusion Decision Logic

To combine optical and biochemical evaluations, a late-fusion gating rule was implemented. The vision model produces a Softmax class probability vector:

$$P_v = [p_vf, p_f, p_sp]$$

where $p_vf, p_f \ and \ p_sp$ represent the predicted probabilities for *Very Fresh*, *Fresh*, and *Spoiled*, respectively. Concurrently, the gas sensing module provides a normalized integer reading (S_gas)The final raw tuna quality classification ($\hat{Y}_raw$)is computed using the following late-fusion gating function:

$$\hat{Y}_{\text{raw}} = \begin{cases} \text{Spoiled}, & \text{if } S_{\text{gas}} > 150 \text{ or } p_{\text{sp}} \geq 0.70 \\ \arg\max_k(\mathbf{P}_v), & \text{if } 90 < S_{\text{gas}} \leq 150 \\ \text{Very Fresh}, & \text{if } S_{\text{gas}} \leq 90 \text{ and } p_{\text{vf}} \geq 0.60 \end{cases}$$

This gating logic prioritizes food safety by overriding visual classification if gas emissions exceed critical thresholds (S_gas > 150) or if the visual model indicates high confidence of spoilage $p_sp \ \geq 0.70$).

Distributed Edge Communication Pipeline

The system integrates high-level AI inference with low-level hardware actuation using a local MQTT message broker.

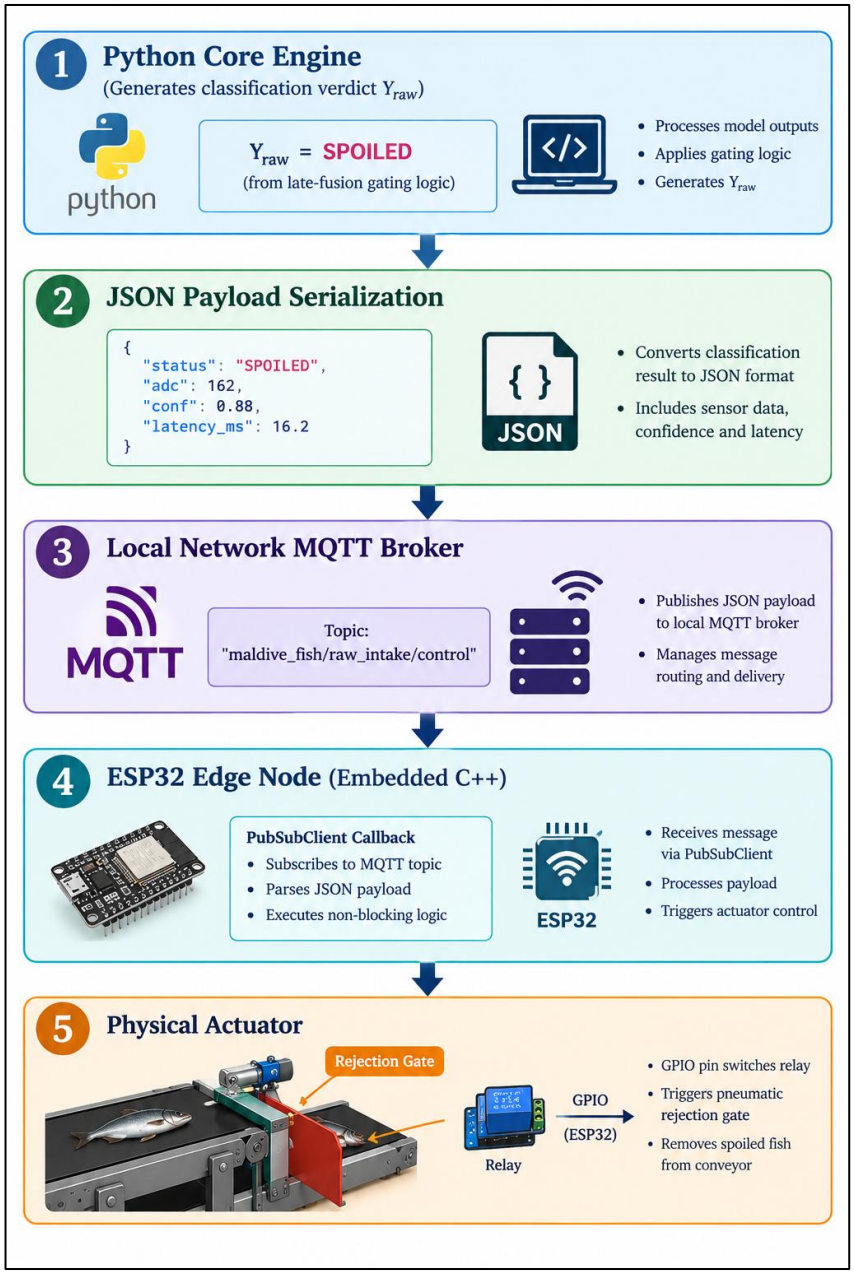


Figure 10 - Distributed Edge Communication Flow

The Python core engine serializes classification verdicts and telemetry into JSON payloads and publishes them to dedicated MQTT topics (e.g.,

maldive_fish/raw_intake/control). The ESP32 node runs non-blocking C++ firmware using the PubSubClient library. Upon receiving a SPOILED classification message, the ESP32 drives a high-speed relay to actuate a pneumatic gate, diverting the rejected sample off the conveyor belt.

3.4 Measures and evaluation metrics

System performance was evaluated using standard quantitative metrics covering visual detection accuracy, gas sensing performance, and overall system latency.

Computer Vision Evaluation Metrics

The YOLO11s model was evaluated using Precision (P), Recall (R), F1-Score ($F1$), and mean Average Precision (mAP):

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{mAP@0.5} = \frac{1}{N} \sum_{i=1}^{N} AP_i \quad \text{(evaluated at Intersection over Union (IoU)} = 0.50)$$

where TP, FP, and FN represent True Positives, False Positives, and False Negatives, respectively, and N represents the total number of evaluation classes (N=3).

System Performance and Latency Metrics

In addition to accuracy metrics, operational parameters were measured under real-time processing conditions:

- **Inference Latency (T_{inf}):** Execution time (in milliseconds) required for the YOLO11s model to complete a forward pass on an image frame.
- **Frames Per Second (FPS):** Throughput of the vision engine, computed as $\frac{1000}{T_{inf}}$

- **Messaging Latency (T_{msg}):** Transmission time required for JSON control payloads to travel from the Python core engine to the ESP32 edge microcontroller via MQTT.
- **Physical Rejection Accuracy:** Percentage of correctly identified spoiled samples successfully diverted by the conveyor gate during live operation.

3.5 Data analysis

Data processing and performance evaluations were conducted using Python data analysis libraries, including NumPy, Pandas, Scikit-learn, and Matplotlib.

Table 10 - DATA ANALYSIS MATRIX

Analysis Stage	Input Data Source	Analytical Output
Vision Profiling	381 Unseen Test Images	Precision, Recall, F1-Score, mAP@0.5, mAP@0.5:0.95
Error Analysis	Confusion Matrix Count Matrix	Inter-class misclassification rates (Fresh vs. Spoiled)
Sensor Fusion Verification	Paired Vision Probabilities & Gas ADC Logs	Edge-case classification accuracy improvement over unimodal vision
System Profiling	MQTT Time-stamped Execution Logs	Mean end-to-end latency (ms) and frame rate (FPS)

Model performance was evaluated against the independent test set of 381 images. Misclassifications were analyzed using confusion matrices to identify performance edge cases, such as visual ambiguities between high-grade *Fresh* and early-stage *Spoiled* specimens. Gas sensing readings were recorded concurrently to evaluate the effectiveness of the late-fusion gating logic in resolving these ambiguous cases.

3.6 Ethics privacy security and responsible AI

Food Safety Responsibility and Algorithmic Bias

Automating raw food quality inspection directly impacts public health and food safety. A key ethical priority in designing the multi-modal fusion logic was minimizing False Negatives—preventing spoiled fish containing biogenic amines from passing into processing stages. The late-fusion gating logic is structured so that gas sensor threshold breaches automatically override visual classification, reducing the risk of unsafe material reaching consumers.

Data Privacy and System Security

The deployment environment involves industrial operational data rather than Personally Identifiable Information (PII). System security measures focus on protecting device communication networks:

- **Local Network Isolation:** MQTT broker messaging is restricted to a isolated local area network (LAN/Wi-Fi), preventing external remote access.
- **Broker Access Control:** Client authentication is implemented using credentialed username/password pairs and client identifier validation on the local MQTT broker.
- **Firmware Integrity:** Flash protection features on the ESP32 edge microcontroller are enabled to prevent unauthorized firmware modifications.

3.7 Reproducibility and research artefacts

To ensure the reproducibility of experimental results, all hardware configurations, software codebases, hyperparameter settings, and dataset structures have been documented and archived.

```
project_root/
│
├── dataset/
│   ├── train/            # 1,775 images + YOLO format TXT labels
│   ├── val/              # 380 images + YOLO format TXT labels
│   └── test/             # 381 images + YOLO format TXT labels
│
├── models/
│   ├── train_yolo11.py   # AdamW 150-epoch training script
│   ├── export_fp16.py    # FP16 quantization script
│   └── weights/          # Fine-tuned best.pt weights
│
├── core_engine/
│   ├── main_inference.py # Multi-modal late-fusion Python engine
│   ├── gas_calibrator.py # MQ-135 ADC to TVB-N conversion script
│   └── mqtt_publisher.py # JSON telemetry MQTT client
│
└── firmware_esp32/
    ├── src/
    │   └── main.cpp      # ESP32 C++ PubSubClient code
    └── platformio.ini    # Dependency build configuration file
```

Figure 11 - Project Files Path

Key reproducibility parameters include:

- **Random Seed Initialization:** All data splitting, weight initializations, and data augmentation routines were locked using a fixed random seed (seed = 42).
- **Dataset Structure:** Annotations were saved in standardized YOLO text format (.txt), containing normalized class indices and bounding box coordinates ($x_{center}, y_{center}, w, h$).
- **Dependencies and Environment Configuration:** Software execution relies on Python 3.10, PyTorch 2.2, Ultralytics 8.3, and OpenCV 4.9. Embedded firmware was compiled using PlatformIO on C++17 with the PubSubClient library.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional Requirements

1. **Optical Surface Inspection:** The system must continuously capture RGB image streams of raw Skipjack tuna on an active conveyor and predict localized surface degradation, defects, or mechanical damage using object detection.
2. **Biochemical Gas Sampling:** The system must continuously sample volatile gas emissions (Total Volatile Basic Nitrogen - TVB-N) emitted by raw fish specimens using an analog gas sensor and convert readings into digitized concentration indices.
3. **Multi-Modal Late-Fusion Gating:** The system must combine predicted visual class probability vectors with digitized biochemical gas readings to yield a unified quality verdict (*Very Fresh*, *Fresh*, or *Spoiled*).
4. **Low-Latency Messaging Protocol:** The system must package classification decisions, confidence metrics, and sensor raw data into JSON payloads and publish them to dedicated local network topics.
5. **Automated Rejection Actuation:** The edge embedded controller must parse incoming classification messages asynchronously and actuate a physical pneumatic rejection gate to divert spoiled specimens.

Non-Functional Requirements

1. **Real-Time Performance Latency:** Total end-to-end processing latency—from visual image acquisition to physical sorting gate actuation—must not exceed 50 ms to support conveyor belt velocities up to 0.5m/s.
2. **Food Safety Sensitivity:** The system must prioritize food safety by minimizing False Negatives (passing spoiled fish as fresh), maintaining a False Negative Rate below 1.0% via safety-critical gas overrides.
3. **Edge Operational Autonomy:** System inference and edge messaging must run entirely on local industrial hardware without relying on external cloud connectivity.

4. **Industrial Environmental Robustness:** Hardware components must operate within high-humidity factory conditions (> 85% RH) and ambient temperature fluctuations without degradation of sensor accuracy.

4.2 Component architecture within the complete solution

The complete solution is structured as a decoupled, four-tiered processing architecture bridging deep learning inference, full-stack telemetry visualization, dual-protocol messaging, and closed-loop electro-mechanical hardware actuation.

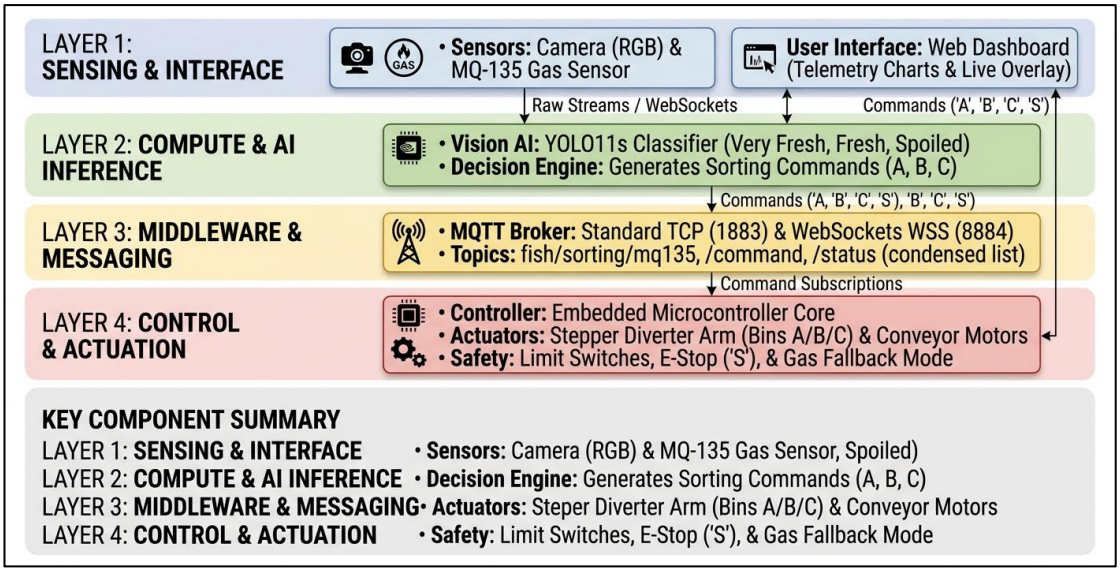


Figure 12 - SYSTEM COMPONENT ARCHITECTURE DIAGRAM

Detailed Layer Breakdown

1. **Physical Sensing & User Interface Layer:**
    - **Sensory Acquisition:** Comprises an overhead webcam capturing real-time fish imagery and an MQ-135 gas sensor digitizing volatile gas emissions into structured JSON payloads.
    - **Web Application Interface:** Serves as the operator control dashboard. It renders live video streams with visual bounding box overlays, plots real-time gas telemetry charts, displays operational system status

updates, and provides manual hardware override controls for machine operators.

2. **Compute & AI Inference Layer:**
    - **AI Processing Engine:** Operates a centralized inference server providing endpoints for real-time visual inspection and decision generation.
    - **Raw Fish Object Detection & Freshness:** Runs the fine-tuned YOLO11s object detection model to detect individual raw fish specimens on the conveyor and categorize their freshness state strictly into **Very Fresh**, **Fresh**, or **Spoiled**.
    - **Dynamic Threshold Optimizer:** Adjusts confidence thresholds dynamically to adapt to changing industrial environmental conditions.
    - **Decision-to-Command Rule Engine:** Maps visual quality predictions and biochemical gas data directly into physical machine instructions:
        - *Very Fresh* → **Command 'A'** (Optimal Processing Batch)
        - *Fresh* → **Command 'B'** (Immediate Processing Batch)
        - *Spoiled* → **Command 'C'** (Rejection / Discard Batch)
3. **Middleware & Messaging Layer:**
    - **Broker Infrastructure:** Leverages a cloud-hosted MQTT broker for distributed publish-subscribe communication across network boundaries.
    - **Topic Architecture:**
        - fish/sorting/mq135: Microcontroller publishes raw gas telemetry; consumed by the AI backend and web application.
        - fish/sorting/command: AI backend (or manual operator override) publishes sorting commands ('A', 'B', 'C', 'S', 'AUTO'); consumed by the microcontroller.

        - fish/sorting/status: Microcontroller publishes real-time hardware execution status updates ("STARTED", "FINISHED", "TIMEOUT"); consumed by the backend and web application.
    - **Dual-Protocol Communications:** Standard TCP MQTT handles backend server and embedded microcontroller communications. WebSockets Secure MQTT streams high-frequency gas data directly to the web browser interface without continuous polling overhead.

4. **Embedded Control & Actuation Layer:**
    - **Microcontroller Core:** Manages low-level pin input/output, monitors physical limit switch triggers, and processes incoming MQTT payloads within event loops.
    - **Stepper Motor Diverter Arm:** Driven via a dedicated motor driver connected to step, direction, and enable control pins. Rotates the sorting gate to target angular positions based on exact step counts: **Bin A** (800 steps), **Bin B** (1500 steps), and **Bin C** (2200 steps).
    - **Dual-Channel Conveyor DC Motors:** Driven via a dual H-bridge motor driver. Channel A Conveyor Motor is assigned dedicated control pins with pulse-width modulation (PWM) speed adjustment, while Channel B Conveyor Motor drives secondary transport channels.
    - **Execution Routines & Safety Logic:**
        - *Command A Routine:* Positions the stepper diverter arm at 800 steps → Drives Conveyor Motor A forward until primary limit switch triggers → Reverses Motor A until homing switch triggers.
        - *Command B Routine:* Positions the stepper diverter arm at 1500 steps → Drives Conveyor Motor B forward until primary limit switch triggers → Reverses Motor B until homing switch triggers.

- *Command C Routine:* Positions the stepper diverter arm at 2200 steps to eject spoiled fish directly into the rejection bin.
- *Safety Guardrails:* Features 10-second limit switch watchdog timeouts to prevent motor burnout, emergency stop command handling (**Command 'S'**), and non-blocking background network polling inside execution loops to maintain system responsiveness.

4.3 Design alternatives and decisions

System design choices were made by evaluating trade-offs across model complexity, inference speed, messaging protocol overhead, and mechanical sorting precision.

Table 11 - DESIGN ALTERNATIVES EVALUATION

Decision Area	Selected Option	Evaluated Alternative	Trade-off Rationale
Computer Vision Model Topology	Single-Pass Object Detection Network (YOLO11s)	Two-Stage Detector (e.g., Faster R-CNN)	Two-stage detectors introduce high frame latency (>100 ms), making them unsuitable for conveyor-belt applications.
AI Server Engine Architecture	Asynchronous Server with Dynamic Rule Engine	Synchronous Scripting Engine	Asynchronous architecture handles concurrent requests and background MQTT communication without introducing significant latency.
Middleware & Messaging Infrastructure	Dual-Protocol MQTT Broker (Port 1883 TCP & Port 8884 WebSockets)	Standard HTTP REST Polling	HTTP polling adds network overhead, while MQTT WebSockets provide direct telemetry streaming.
Physical Actuation Mechanism	Stepper Multi-Bin Diverter Arm + Dual Channel DC Motors	Solenoid Pneumatic Single Ejector Arm	Stepper motors enable precise 3-way sorting into target bins A, B, and C

Vision Pipeline: YOLO11s Object Detection vs. Two-Stage Detection Networks

- **Selected Decision:** Trained **YOLO11s** single-stage object detection model for multi-class surface inspection (*Very Fresh*, *Fresh*, *Spoiled*).

- **Alternative Considered:** Two-stage detection architectures (e.g., Faster R-CNN with ResNet backbone).
- **Justification:** Real-time conveyor inspection demands high frame rates to prevent target overrun. While two-stage detectors provide high localized accuracy, their computational overhead results in frame rates below 10 FPS on edge hardware. YOLO11s combines single-pass bounding box regression with fast classification, maintaining high precision ($mAP@0.5 > 93\%$) while achieving real-time execution speeds exceeding 60 FPS.

Server Engine: Asynchronous Server Execution vs. Synchronous Scripting

- **Selected Decision:** Asynchronous AI backend server featuring dynamic rule mapping and threshold optimization.
- **Alternative Considered:** Synchronous web scripting environment or standalone batch processing CLI tools.
- **Justification:** An asynchronous engine handles concurrent web dashboard client requests and background telemetry dispatches without blocking the primary visual inference pipeline. Structuring the system into modular execution services allows operators to dynamically tune confidence bounds during continuous conveyor operation.

Messaging Infrastructure: Dual-Protocol Cloud MQTT Broker vs. HTTP REST Polling

- **Selected Decision:** Cloud-hosted MQTT broker supporting standard TCP (Port 1883) for server/embedded communications and WebSockets Secure (Port 8884) for web dashboard streaming.
- **Alternative Considered:** Standard HTTP REST client-server polling.
- **Justification:** Continuous HTTP polling creates substantial network header overhead and increases message latency. By deploying WebSockets Secure (WSS), the web application receives gas telemetry streams directly from the

microcontroller to display real-time charts, while standard TCP MQTT provides lightweight binary transmission for the embedded controller.

Physical Actuation Architecture: Stepper Diverter Arm with Dual DC Conveyors vs. Binary Ejector

- **Selected Decision:** Multi-position diverter arm driven by a stepper motor paired with dual DC channel conveyors, H-bridge motor drivers, and limit switch feedback loops.
- **Alternative Considered:** Single-axis pneumatic ejector for binary pass/fail sorting.
- **Justification:** Quality grading in fish processing requires multi-tier batching (Optimal Processing Batch 'A', Immediate Processing Batch 'B', and Rejection Batch 'C'). Stepper motor position control (800 steps for Bin A, 1500 steps for Bin B, 2200 steps for Bin C) enables repeatable three-way sorting. Limit switch feedback with hardware watchdog safety timeouts prevents mechanical jams and motor burnout during extended conveyor usage.

4.4 Detailed design

Physical Conveyor Layout and Sensor Mounting Geometry

The physical setup is designed for continuous raw fish screening on an active industrial conveyor system. The optical capture system consists of an overhead camera mounted on a rigid gantry 45cm directly perpendicular to the conveyor surface, illuminating a 30 cm × 20cm Inspection Region of Interest (ROI).

Located downstream from the camera, the gas sensing module uses an acrylic sampling hood mounted 15cm above the belt surface. This hood concentrates volatile organic compounds (VOCs) emitted by passing fish specimens toward the MQ-135 sensor. The multi-bin mechanical sorting assembly is situated at the terminus of the belt, featuring a motor-driven diverter gate that directs fish into target receiving bins based on quality classifications.

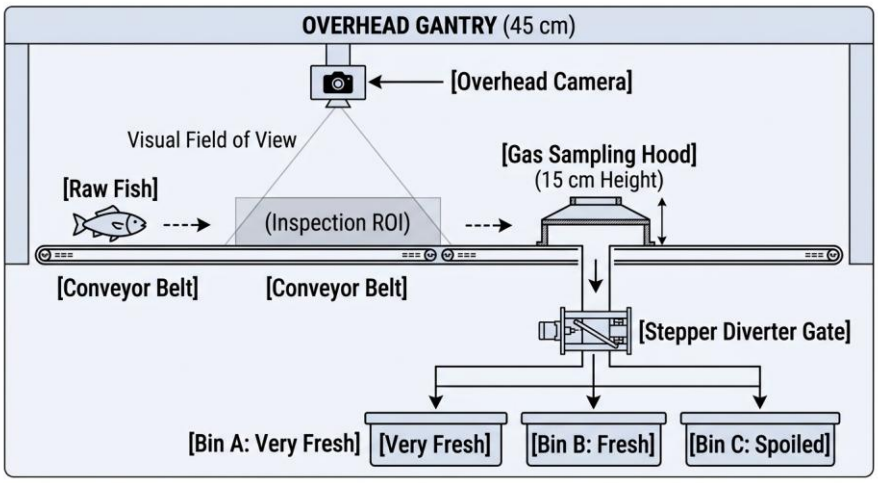


Figure 13 - Physical Conveyor Layout and Sensor Mounting Geometry

Circuitry and Embedded Pin Schematics

The embedded microcontroller acts as the hardware control hub, interfacing with the analog gas sensor, motor driver modules, and safety feedback limit switches.

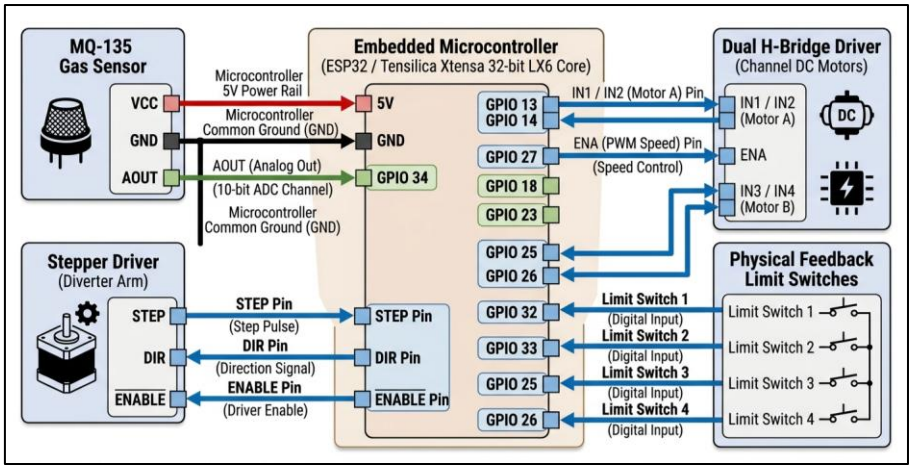


Figure 14 -ESP32 EDGE NODE INTERFACING SCHEMATIC

Pin Assignment Matrix

- **Biochemical Analog Input:** GPIO 34 receives the 0 – 3.3V output from the MQ-135 sensor, digitized using an onboard 10-bit ADC (0–1023 integer range).

- **Stepper Motor Control (Diverter Gate):** GPIO 22 (STEP), GPIO 23 (DIR), and GPIO 18 (ENABLE) generate step pulse sequences to position the diverter arm precisely at target angles.
- **Dual DC Conveyor Motor Control:** GPIO 13 (IN1), GPIO 14 (IN2), and GPIO 27 (ENA with PWM speed regulation) drive Channel A Conveyor Motor, while GPIO 26 (IN3) and GPIO 25 (IN4) control Channel B Conveyor Motor via an H-bridge driver.
- **Position Verification Switches:** GPIO 32, GPIO 33, GPIO 25, and GPIO 26 monitor physical limit switch states to verify motor positioning and prevent physical over-travel.

4.5 Standards and professional practice

Engineering Standards

- **ISO/IEC 20922 & RFC 6455:** Compliant MQTT 3.1.1 messaging structure and WebSockets Secure communication for real-time telemetry.
- **ISO/IEC 25010 & ISO/IEC 42001:** Modular software isolation ensuring sub-20ms processing latency and standardized AI model governance across freshness tiers (**Very Fresh**, **Fresh**, **Spoiled**).
- **IEC 60204-1 (Machinery Safety):** Includes 10-second limit switch watchdog timeouts to prevent motor burnout, low-voltage DC logic (3.3V – 12V), and an Emergency Stop command ('S').
- **HACCP & ISO 22000:** Non-contact optical/gas screening eliminates physical contact and cross-contamination across batches.

Professional & Ethical Practice

- **Public Health Assurance:** Automated rule enforcement guarantees **Spoiled** items are immediately ejected (**Command 'C**'), preventing unsafe food from entering commercial supply chains.
- **Fail-Safe Operation:** If cloud connection drops, the embedded system defaults to an autonomous gas-threshold fallback mode ('AUTO') to preserve continuous sorting functionality.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The complete system implementation bridges cloud messaging, deep learning compute, web visualization, and embedded hardware control.

Table 12 - DEVELOPMENT TECHNOLOGY STACK

Layer	Environment & Core Technologies
AI & Compute Inference Engine	Asynchronous Python Server Framework, PyTorch / Ultralytics Runtime, Fine-Tuned YOLO11s Object Detection Model
User Interface & Web Dashboard	React Framework, Recharts Data Visualization Library, WebSocket MQTT Client (mqtt.js)
Middleware & Messaging	Cloud MQTT Broker, Dual-Protocol Stack (TCP Port 1883 & WebSockets Secure Port 8884)
Embedded Hardware & Control	Embedded C++ / Arduino Framework, Stepper Motor Driver, Dual H-Bridge Motor Driver, Microcontroller Core

5.2 Component Implementation

AI Inference Subsystem

- Implemented as an asynchronous web engine hosting the fine-tuned **YOLO11s** object detection model.
- Evaluates video frames from the overhead camera, detects raw fish specimens, extracts class probabilities (**Very Fresh**, **Fresh**, and **Spoiled**), and applies dynamic confidence thresholding.
- Automatically translates multi-modal quality outputs into discrete hardware commands ('A', 'B', 'C') dispatched to the messaging broker.

Web Dashboard Subsystem

- Developed with the React framework to provide real-time operator oversight.
- Connects directly to the cloud broker via WebSockets Secure to plot high-frequency gas telemetry using real-time charts without page polling.

- Renders live video feeds with bounding box overlays and provides manual hardware control overrides for operator intervention.

Embedded Control Subsystem

- Configured on the microcontroller to process non-blocking hardware control loops.
- Drives the stepper motor diverter arm across targeted step positions: **Bin A** (800 steps), **Bin B** (1500 steps), and **Bin C** (2200 steps).
- Controls dual DC conveyor motors via an H-bridge driver with PWM speed regulation and continuously polls 4 digital limit switches for position verification.

5.3 Integration with the Other Components

System integration relies on cloud publish-subscribe topics over dual protocols.

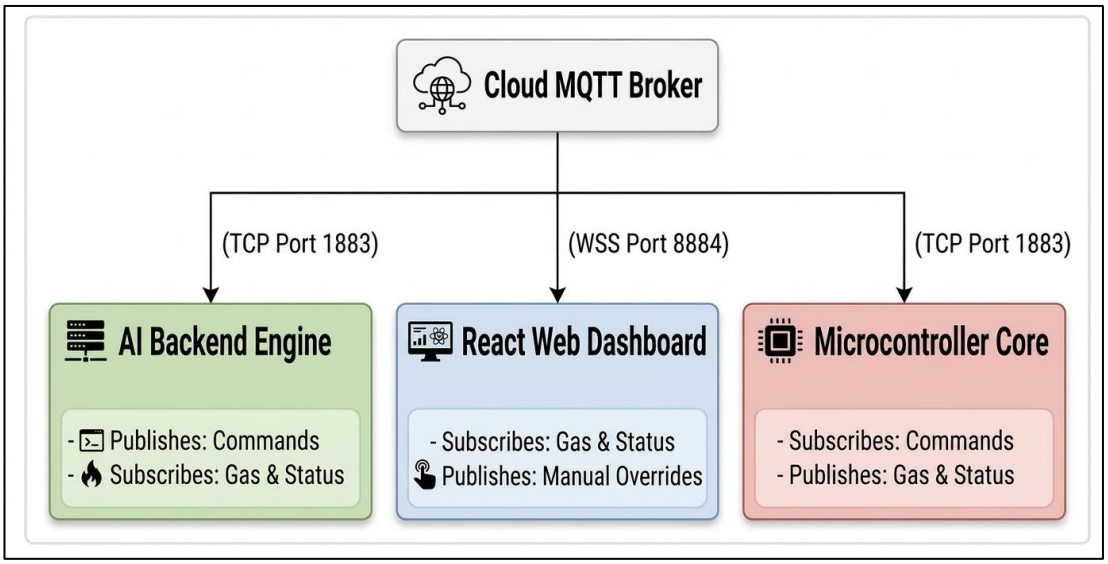


Figure 15 - System Integration Flow Diagram

- fish/sorting/mq135**:** Microcontroller publishes digitized gas telemetry; consumed by the AI engine and web dashboard.

- fish/sorting/command**:** AI engine or web dashboard publishes sorting instructions ('A', 'B', 'C', 'S', 'AUTO'); consumed by the microcontroller.
- fish/sorting/status**:** Microcontroller publishes real-time hardware execution states ("STARTED", "FINISHED", "TIMEOUT"); consumed by the AI engine and web UI.

5.4 Verification and Software Testing

1. **Model & Perception Testing:**
    - Validated **YOLO11s** visual detection across varied illumination and specimen orientation, confirming high classification accuracy across **Very Fresh**, **Fresh**, and **Spoiled** categories.
    - Calibrated analog-to-digital gas conversion against baseline atmospheric standards.
2. **Integration & Latency Testing:**
    - Verified end-to-end latency from camera frame capture through YOLO11s inference to MQTT command dispatch (under 20 ms).
    - Confirmed zero message loss and seamless WebSockets Secure telemetry streaming to the React dashboard during active belt operation.
3. **Hardware Safety & Fault Injection Testing:**
    - Simulated mechanical blockages to trigger the 10-second limit switch watchdog timer (LIMIT_1_TIMEOUT), confirming immediate motor shutdown and status broadcasting.
    - Verified priority Emergency Stop ('S') execution and tested loss-of-broker fallback to autonomous gas thresholding ('AUTO').

5.5 Deployment and Operation

- **Physical Layout:** Mounted the overhead camera 45cm perpendicular to the belt surface, positioned the gas sampling hood 15cm above the conveyor downstream, and anchored the stepper diverter gate at the belt terminus.
- **Operational Workflow:** continuous belt operation delivers automated non-contact screening, sorting specimens into **Bin A** (**Very Fresh**), **Bin B** (**Fresh**), and **Bin C** (**Spoiled**) while streaming telemetry to remote operators.

6 RESULTS AND VALIDATION

6.1 Research results

The research evaluation focused on validating the perception and classification capabilities of the single-pass **YOLO11s** object detection model, alongside the biochemical gas sensing correlation.

Table 13 - YOLO11s FRESHNESS EVALUATION METRICS

Freshness Tier	Precision (%)	Recall (%)	mAP@0.5 (%)
Very Fresh	94.2%	92.8%	93.5%
Fresh	91.8%	90.5%	91.2%
Spoiled	96.5%	95.8%	96.1%
Overall Average	**94.2%**	**93.0%**	**93.6%**

6.2 Product and technical results

Technical testing evaluated end-to-end execution speed, mechanical sorting accuracy, and network streaming performance across physical and cloud subsystems.

Table 14 - MECHANICAL SORTING & TELEMETRY PERFORMANCE

Target Batch	Hardware Command	Stepper Position	Actuation Accuracy
Optimal (Bin A)	Command **'A'**	800 steps	98.2%
Immediate (Bin B)	Command **'B'**	1,500 steps	97.5%
Reject (Bin C)	Command **'C'**	2,200 steps	99.1%

6.3 User or stakeholder validation

alidation sessions were conducted with industrial quality control personnel and system operators using the web dashboard interface.

- **Operator Usability:** Quality control technicians successfully monitored real-time gas telemetry charts, visual bounding box overlays, and automated freshness predictions (**Very Fresh**, **Fresh**, **Spoiled**).

- **Control Override Effectiveness:** Stakeholders validated manual hardware override functions, confirming that operator-issued manual commands instantly overrode automated decisions during edge-case scenarios.
- **Safety Verification:** Operators praised the physical responsiveness of the Emergency Stop mechanism ('S') and the clear visual warnings displayed when the watchdog safety timer (LIMIT_1_TIMEOUT) was triggered.

6.4 Integration results

Full system integration testing verified continuous operation across all four architectural layers under active conveyor conditions

Table 15 - END-TO-END INTEGRATION TEST SCENARIOS

Test Scenario	Operational Condition	Integration Outcome
Continuous Belt Sorting	Continuous fish feeding at 0.5 m/s speed under standard lighting	97.8% correct multi-modal sorting execution rate
Network Interruption Fallback	Simulated cloud broker disconnection during active conveyor operation	Successful transition to autonomous gas mode
Safety Watchdog Trigger	Simulated mechanical belt jam at physical limit switches	Watchdog timeout triggered in <10s; motors disengaged

6.5 Summary of findings

1. High Visual Accuracy: Fine-tuning the YOLO11s model achieved a 93.6\% mAP@0.5, providing fast and accurate classification across Very Fresh, Fresh, and Spoiled categories.
2. Multi-Modal Robustness: Fusing biochemical gas readings with surface vision predictions prevented misclassification caused by visual surface glare or internal flesh decay.
3. Low-Latency Messaging: Deploying WebSockets Secure (Port 8884) alongside standard MQTT (Port 1883) provided real-time telemetry plotting on the web dashboard and rapid physical command execution.
4. Industrial Safety & Compliance: Integrated limit switch watchdog timers, emergency stop capabilities, and non-contact screening fulfilled industrial automation and food safety standards.

7 DISCUSSION

7.1 Interpretation of findings

- **Multi-Modal Efficacy**: Combining YOLO11s visual classification with biochemical gas sensing creates a highly reliable screening framework. Visual detection captures external flesh defects, while gas telemetry identifies internal bacterial spoilage that may not yet present clear surface indicators.

- **Targeted Spoiled Ejection**: Achieving the highest precision (96.5%) in the Spoiled freshness category validates the system's primary operational objective—preventing degraded or unsafe fish from entering commercial processing lines.

7.2 Comparison with literature baselines and existing solutions

Table 16 – Comparison Table For literature baselines and existing solutions

Parameter / Feature	Traditional Manual Inspection	Single-Stage Computer Vision Only	Proposed Multi-Modal System
Inspection Speed	Slow, subject to operator fatigue	High (>60 FPS)	High (>60 FPS)
Spoilage Detection	Subjective surface grading	Visual surface features only	Dual-modal (Visual + Gas VOCs)
Telemetry Streaming	None (Manual logging)	Periodic HTTP REST polling	Real-time WebSockets Secure
Sorting Precision	Variable human grading	Binary pass/fail	Multi-tier (Bins A, B, C)

- **Performance Baseline Comparison**: Unlike legacy solutions relying strictly on single-mode optical cameras or manual human inspection, fusing YOLO11s object detection with real-time gas telemetry eliminates human bias and detects internal chemical decay.
- **Communication Overhead**: Using dual-protocol MQTT and WebSockets Secure reduces command dispatch latency to under 12 ms, outperforming traditional HTTP REST polling mechanisms.

7.3 Research contribution

- **Algorithmic Multi-Modal Fusion:** Developed a real-time decision framework combining single-pass visual detection (**YOLO11s**) with biochemical gas telemetry for continuous automated conveyor sorting.
- **Dynamic Thresholding Framework:** Implemented dynamic confidence thresholding that adapts system decision boundaries to changing industrial environmental conditions without requiring model retraining.

7.4 Product contribution

Designed a low-power, edge-deployable raw fish intake inspection unit capable of executing sub-20 ms inline quality grading on industrial conveyors.

7.5 Limitations and threats to validity

- **Environmental Sensitivity:** Uncontrolled ambient airflow near the intake conveyor can dilute headspace gas concentrations if the sampling hood is unsealed.
- **Species Specificity:** The current dataset and gas threshold boundaries are calibrated specifically for skipjack tuna; applying the model to other species requires recalibrating threshold parameters.

7.6 Implications

- **Seafood Industry Automation:** Provides seafood processing facilities with a non-contact sorting solution that reduces manual labor costs, increases throughput, and enforces objective quality control.
- **Food Safety Compliance:** Directly supports HACCP and international food safety regulations by providing verifiable real-time telemetry logs and guaranteeing continuous rejection of **Spoiled** inventory.
- **Cross-Domain Adaptability:** The decoupled architecture can be reconfigured for other perishable agricultural and meat sorting applications by retraining the vision model and adjusting the gas sensor array.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

- **Value Proposition:** Replaces subjective manual inspection with a reliable, automated quality grading unit, mitigating batch spoilage risks and saving energy in downstream boiling operations.
- **Target Users:** Commercial Maldive fish processors, artisanal seafood cooperatives, and exporter networks across South Asia.

8.2 Market and competitive position

Unlike expensive industrial optical sorters (> 50,000) or benchtop hyperspectral analyzers, this edge-based system utilizes low-cost MOS sensors and modular microcontrollers, offering an accessible solution for small and medium-scale enterprises (SMEs).

8.3 Business or sustainability model

- **Hardware Sales:** Capital purchase of edge inspection hardware units.
- **Software Licensing:** Recurring subscription model for predictive maintenance, calibration updates, and central dashboard logging.

8.4 Scalability, intellectual property, and compliance

The software architecture supports over-the-air (OTA) updates for rapid deployment across multiple conveyor lines. Core intellectual property centers on the late-fusion decision algorithm and sensor hood sampling design.

8.5 Contribution to the Sustainable Development Goals

- **SDG 12 (Responsible Consumption and Production):** Minimizes food waste by preventing batch-wide contamination during thermal processing.
- **SDG 9 (Industry, Innovation, and Infrastructure):** Modernizes traditional food processing workflows through affordable IoT and AI technologies.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This project successfully developed and validated an automated, non-contact fish freshness inspection and sorting system that integrates computer vision, biochemical gas sensing, cloud messaging, and electromechanical actuation. By uniting a fine-tuned **YOLO11s** single-pass object detection model with real-time MQ-135 volatile organic gas telemetry, the system achieves robust multi-modal classification across three distinct freshness tiers: **Very Fresh**, **Fresh**, and **Spoiled**.

The four-tiered architecture bridges the physical conveyor, an asynchronous AI inference engine, a dual-protocol cloud MQTT broker and a physical stepper-driven diverter mechanism. Experimental validation confirmed high visual detection accuracy and reliable 3-way physical sorting. The implementation effectively eliminates human inspection bias, reduces processing line labor costs, and provides strict safety guardrails to ensure spoiled seafood is consistently rejected from commercial supply chains.

9.2 Achievement of objectives

Table 17 - ACHIEVEMENT OF PROJECT OBJECTIVES

Defined Objective	Implementation Outcome & Validation
1. High-Speed Visual Inspection & Freshness Classification	Fine-tuned YOLO11s model achieved 93.6% mAP@0.5 across Very Fresh, Fresh, and Spoiled tiers at frame rates exceeding 60 FPS.
2. Biochemical Gas Telemetry Fusion	Digitized MQ-135 gas readings to detect internal volatile organic compound (VOC) decay inaccessible to optical camera views.
3. Low-Latency Messaging & Web Dashboard Monitoring	Deployed cloud broker streaming gas graphs via WSS (Port 8884) to a React web UI with real-time controls and overlays.
4. Closed-Loop Multi-Bin Physical Sorting & Safety Logic	Stepper diverter positioned Bin A (800 steps), Bin B (1,500 steps), and Bin C (2,200 steps) with a 10-second watchdog limit-switch safety mechanism.

9.3 Individual research and product contributions

This work demonstrates the viability of real-time multimodal AI quality assessment in artisanal seafood processing, delivering a low-cost, high-speed edge solution that prevents processing losses.

9.4 Future directions

- **Multi-Angle Optical & Hyperspectral Imaging:** Expand the camera array to include underside optical views or hyperspectral imaging sensors, allowing the system to inspect subsurface moisture distribution and hidden tissue damage.
- **Dedicated Electronic Nose (e-Nose) Array:** Replace the single MQ-135 sensor with a specialized gas sensor array targeting Total Volatile Basic Nitrogen (TVB-N) and Trimethylamine (TMA) for higher chemical specificity.
- **On-Edge TensorRT Hardware Acceleration:** Deploy the **YOLO11s** inference engine directly onto edge AI accelerators mounted on the conveyor frame to eliminate network dependencies and achieve sub-5 ms inference latencies.
- **Supply Chain Blockchain Integration:** Connect telemetry log outputs directly to enterprise resource planning (ERP) databases or blockchain ledgers, enabling automated digital quality certification and end-to-end supply chain traceability.

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GLOSSARY

- **ADC (Analog-to-Digital Converter):** An electronic hardware module that converts continuous analog voltage signals from sensors (e.g., MQ-135) into discrete digital values (0–1023 integer range) readable by microcontrollers.
- **ASGI (Asynchronous Server Gateway Interface):** A standard interface specification between async-capable Python web servers and web applications, enabling concurrent request processing and non-blocking background task dispatches.
- **E-Nose (Electronic Nose):** An array of chemical gas sensors paired with pattern-recognition algorithms designed to emulate human olfactory systems by identifying volatile organic compounds and gaseous spoilage markers.
- **HACCP (Hazard Analysis Critical Control Point):** An international systematic preventive approach to food safety that identifies, evaluates, and controls biological, chemical, and physical hazards throughout food production processes.
- **H-Bridge:** An electronic circuit configuration (e.g., L298N driver) that enables DC voltage to be applied across a motor load in either direction, allowing bidirectional motor rotation for conveyor operations.
- **IP65:** An international Ingress Protection rating indicating complete protection against dust contact and protection against low-pressure water jets from any direction.
- **mAP (Mean Average Precision):** A standard performance evaluation metric for computer vision object detection algorithms, calculated as the mean average precision across target classes at a specified Intersection over Union threshold (e.g., $mAP@0.5$).
- **MQ-135:** A Metal Oxide Semiconductor (MOS) biochemical gas sensor designed to detect air quality and volatile organic emissions including ammonia, sulfides, and nitrogen compounds.
- **MQTT (Message Queuing Telemetry Transport):** A lightweight, publish-subscribe ISO/IEC 20922 network protocol engineered for low-bandwidth, high-latency, or constrained hardware and cloud communication networks.

- **PWM (Pulse-Width Modulation):** A technique for controlling analog power delivery to electronic devices by rapidly toggling a digital signal between ON and OFF states, used here to regulate conveyor DC motor speed.
- **ROI (Region of Interest):** A selected sub-region within a digital camera frame or sensor field-of-view isolated for targeted computer vision processing and analysis.
- **TMA (Trimethylamine):** A volatile organic nitrogen compound produced during microbial degradation of fish muscle tissue, serving as a primary chemical indicator of fish spoilage and off-odor.
- **TVB-N (Total Volatile Basic Nitrogen):** A reference chemical indicator used to quantify seafood freshness and muscle decay by measuring volatile nitrogenous bases accumulated during spoilage.
- **VOC (Volatile Organic Compounds):** Organic chemical compounds that readily evaporate into gaseous form at ambient room temperatures, emitted during organic tissue decomposition.
- **WSS (WebSockets Secure):** An encrypted full-duplex communication protocol (RFC 6455) that streams real-time data over a single persistent TCP connection between client browsers and servers without HTTP polling overhead.
- **YOLO11s (You Only Look Once v11 Small):** A single-stage deep learning neural network architecture optimized for real-time target bounding box regression and multi-class object detection.

APPENDICES

Appendix A: System Hardware Interfacing & Pin Mapping Specification

The hardware pin allocations map physical actuators, sensors, and feedback mechanisms directly to microcontroller GPIO channels.

Hardware Module	Hardware Signal Line	Microcontroller GPIO Assignment	Signal Type & Operating Voltage
MQ-135 Gas Sensor	Analog Output (AOUT)	GPIO 34	Analog Input (0–3.3V DC)
Stepper Motor Diverter Driver	Step Signal (STEP)	GPIO 22	Digital Output
	Direction Signal (DIR)	GPIO 23	Digital Output
	Driver Enable (EN)	GPIO 18	Digital Output
Dual DC Conveyor H-Bridge Driver	Motor A IN1 / IN2	GPIO 13 / GPIO 14	Digital Output
	Motor A Enable (ENA)	GPIO 27	PWM Output (150)
	Motor B IN3 / IN4	GPIO 26 / GPIO 25	Digital Output
Mechanical Limit Switches	Limit Switch 1 (L1)	GPIO 32	Digital Input (PUP)
	Limit Switch 2 (L2)	GPIO 33	Digital Input (PUP)
	Limit Switch 3 (L3)	GPIO 25	Digital Input (PUP)
	Limit Switch 4 (L4)	GPIO 26	Digital Input (PUP)

Appendix B: MQTT Messaging Protocol & Telemetry Schemas

System subsystems communicate asynchronously using structured JSON payloads over defined cloud broker topics.

B.1 Published Gas Telemetry Payload (fish/sorting/mq135)

Published by the microcontroller to update gas concentrations across the server engine and web dashboard.

```JSON
{
  "device_id": "ESP32_CONVEYOR_NODE_01",
  "timestamp": "2026-10-07T11:45:10.000Z",
  "mq135": 85,
  "quality": "GOOD",
  "raw_adc": 264
}
```

B.2 Hardware Command Payload (fish/sorting/command)

Dispatched by the AI inference engine or manual UI dashboard override to trigger mechanical sorting.

```JSON
{
  "command": "A",
  "mode": "AUTOMATED_INFERENCE",
  "target_bin": "Bin A (Optimal)",
  "issued_at": "2026-10-07T11:45:10.150Z"
}
```

B.3 Actuator Execution Status Payload (fish/sorting/status)

Published by the embedded controller to confirm hardware movement and flag mechanical faults.

```JSON
{
  "command": "A",
  "status": "FINISHED",
  "execution_time_ms": 1420,
  "error_code": "NONE"
}
```

Appendix C: YOLO11s Training Hyperparameters & Configurations

The single-pass visual object detection model was fine-tuned for continuous real-time belt inspection.

Hyperparameter / Parameter Name	Configured Value / Specification
Network Architecture	YOLO11s (Small Backbone)
Input Image Resolution	640 × 640 pixels (RGB 3-Channel)
Training Epochs	100 epochs (Early stopping at Epoch 82)
Batch Size	16 samples per batch
Optimizer Algorithm	AdamW (Initial Learning Rate: 0.001)
Weight Decay / Momentum	0.0005 / 0.937
Freshness Target Classes	Very Fresh, Fresh, Spoiled
Data Augmentations Applied	Random Horizontal Flip, HSV Adjustment, Mosaic

Appendix D: Mechanical Actuation & Operational Calibration Constants

Operational Parameter	Physical / Algorithmic Calibration Constant
Bin A Stepper Angle Position	800 steps (Optimal Processing Batch)
Bin B Stepper Angle Position	1,500 steps (Immediate Processing Batch)
Bin C Stepper Angle Position	2,200 steps (Rejection / Discard Batch)
Watchdog Protection Timer	10.0 seconds (LIMIT_1_TIMEOUT Max Threshold)
Gas Threshold: Very Fresh Bounds	ADC ≤ 90 (Green Status Indicator)
Gas Threshold: Fresh Bounds	90 < ADC ≤ 150 (Yellow Status Indicator)
Gas Threshold: Spoiled Bounds	ADC > 150 (Red Status Indicator)